

DENTAL PULP

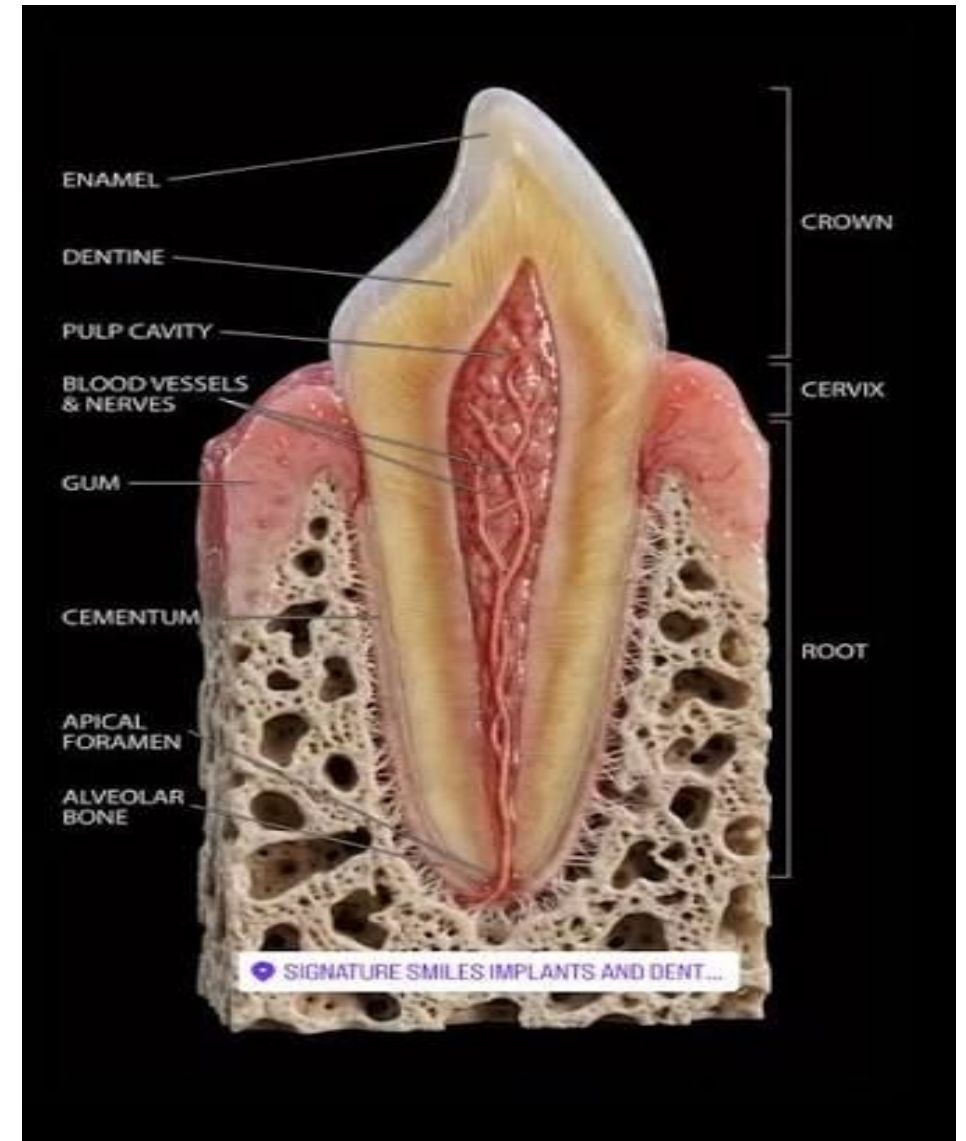
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OVERVIEW

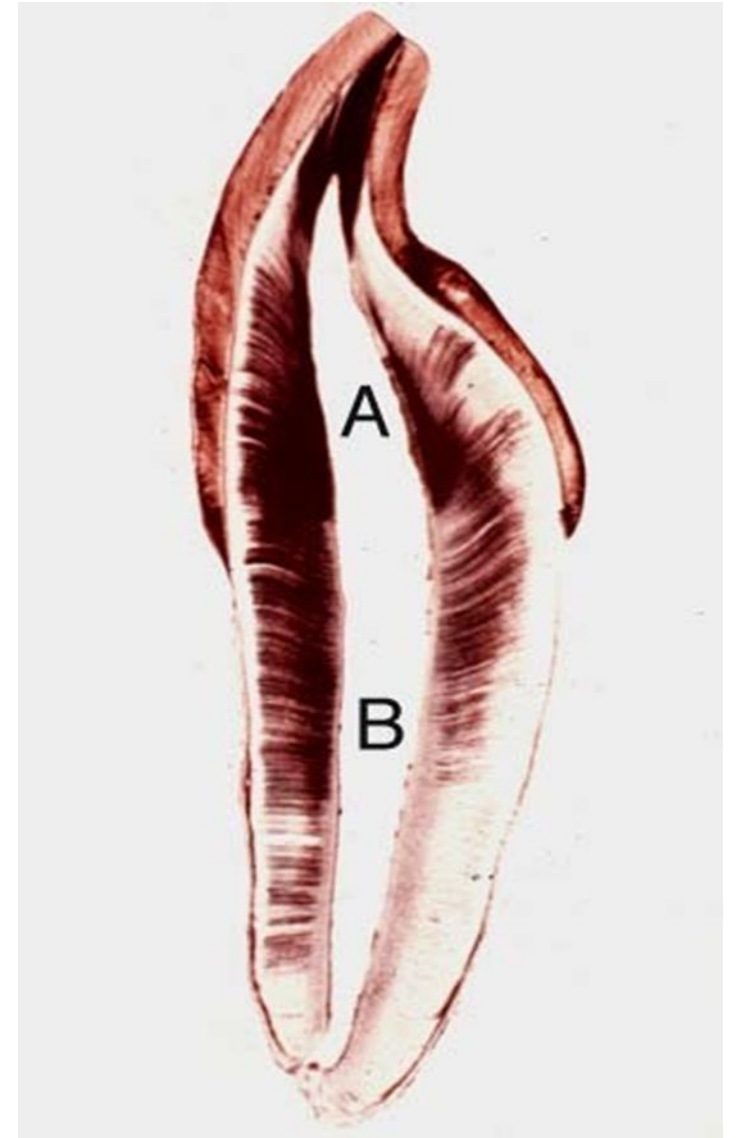
- Dental Pulp is a soft connective tissue derived from **Dental papilla**.
- Occupies the Pulp chamber and root canal.
- At the apical constriction of the root canal it becomes continuous with the periodontal ligament.
- Although the pulp is most obviously active during development and eruption of the tooth, it remains productive throughout life, forming secondary dentine slowly but regularly.



REGIONS OF PULP

- A. CORONAL
- B. RADICULAR

The coronal pulp is located within the crown of the tooth and the radicular pulp is located within the root.



ORGANISATION OF DENTAL PULP

- **Peripheral layer** = odontoblasts (cells responsible for dentine deposition).

Also peripherally located (able to detect external stimuli):

1. **Trigeminal nerve terminals (afferents)**
2. **Dendritic antigen-presenting cell.**

- **Vascular/neural entry:** via apical foramen at root apex.
- Each root has ≥ 1 main canal + apical foramen (many have 2).
- Accessory canals branch off main canal, mostly in apical third of root.
- In multirooted teeth, small vascular canals also enter via interradicular bone.

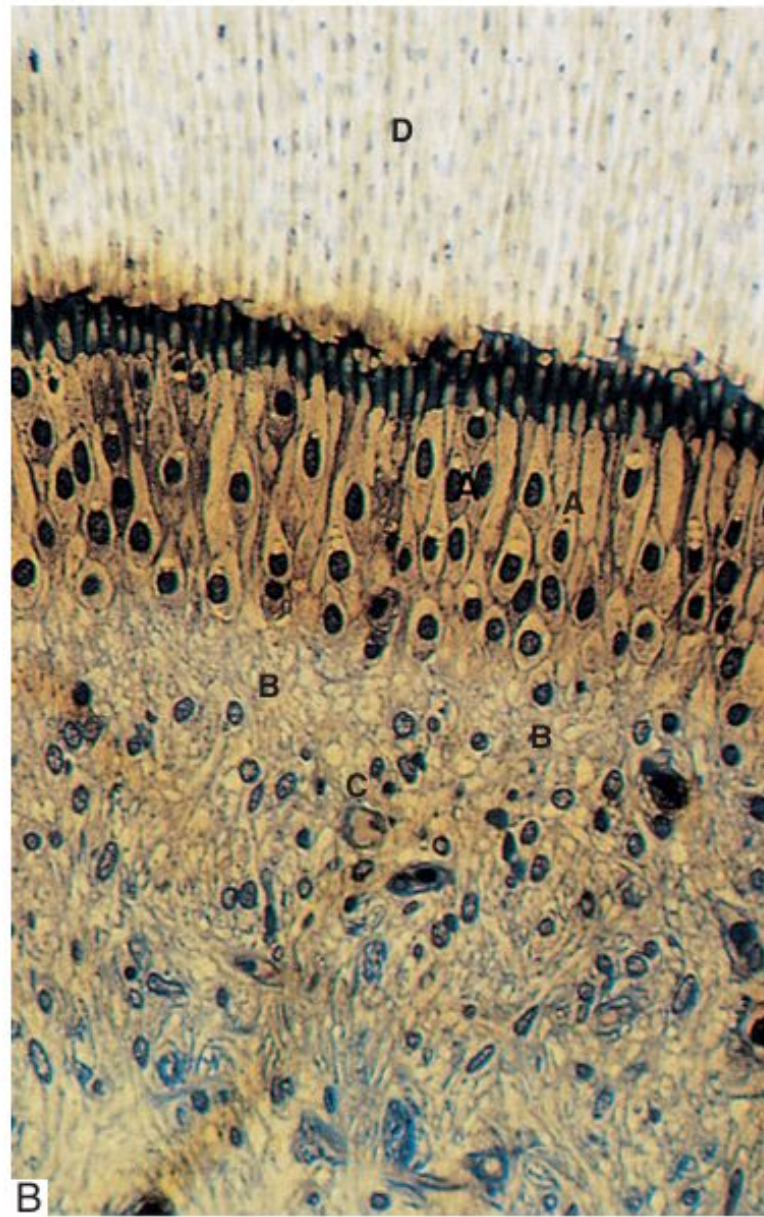
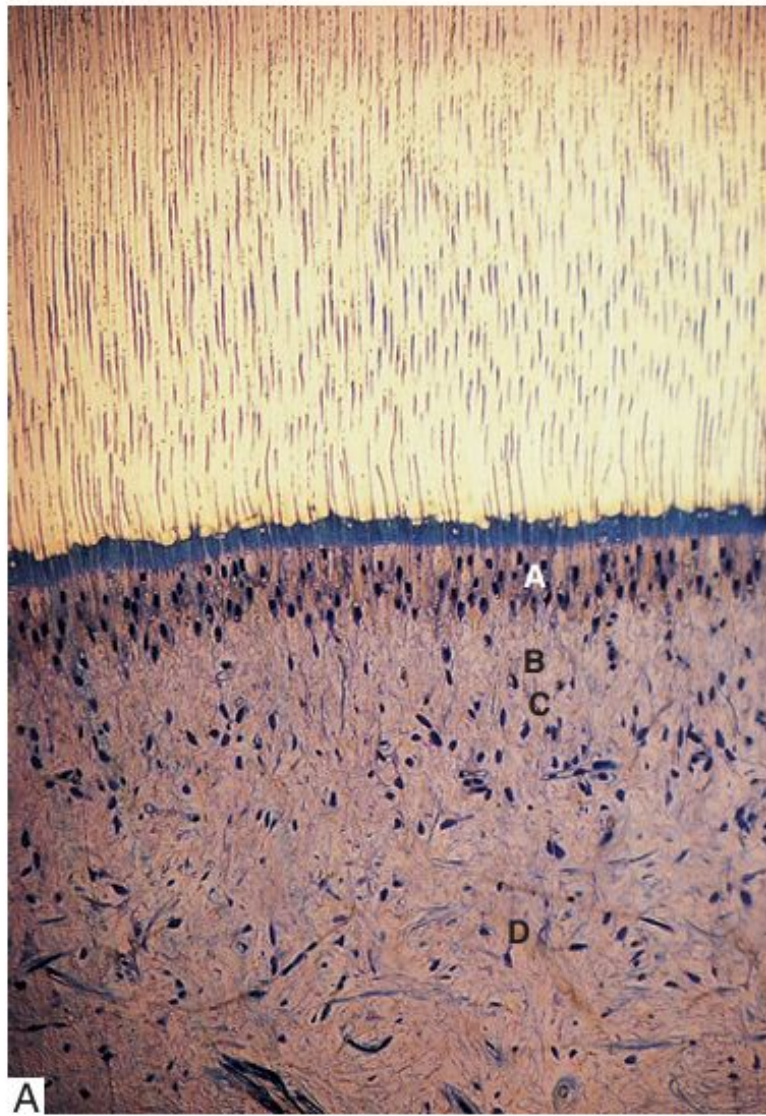


Fig. 10.10 (A) Low-power micrograph showing some cell types and their distribution in the pulp. A = odontoblast layer adjacent to dentine; B = cell-free zone; C = cell-rich zone; D = central pulp showing principally fibroblasts (toluidine blue; $\times 200$). (B) Higher-power view showing pseudostratified odontoblast layer (A). B = cell-free zone; C = cell-rich zone; D = dentine (toluidine blue; $\times 650$).

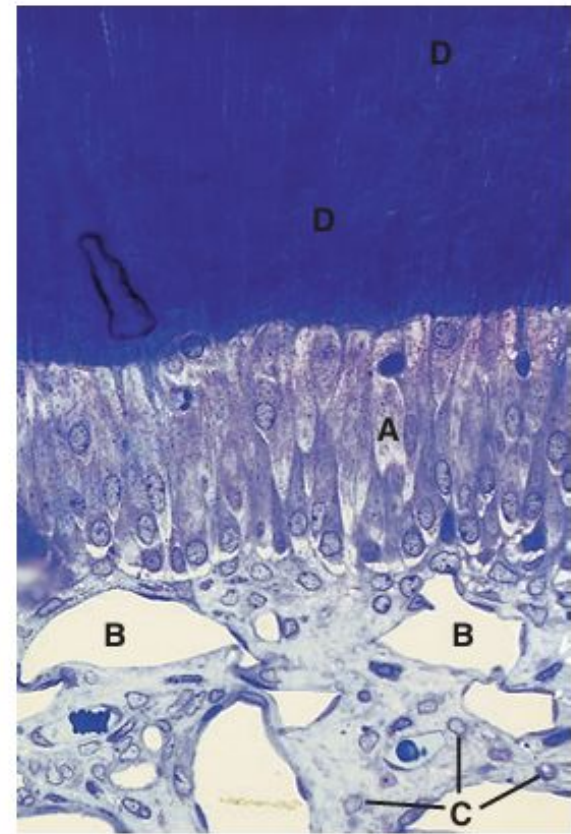


Fig. 10.11 Semi-thin section showing pseudostratified odontoblast layer (A). B = subodontoblastic capillary plexus; C = pulp fibroblasts; D = dentine (toluidine blue; $\times 450$).

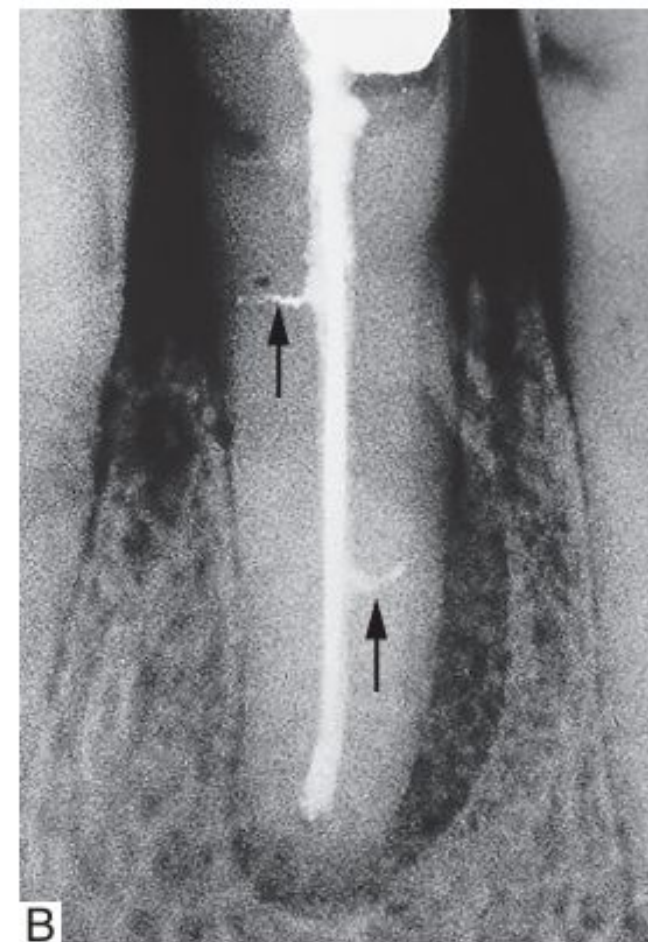
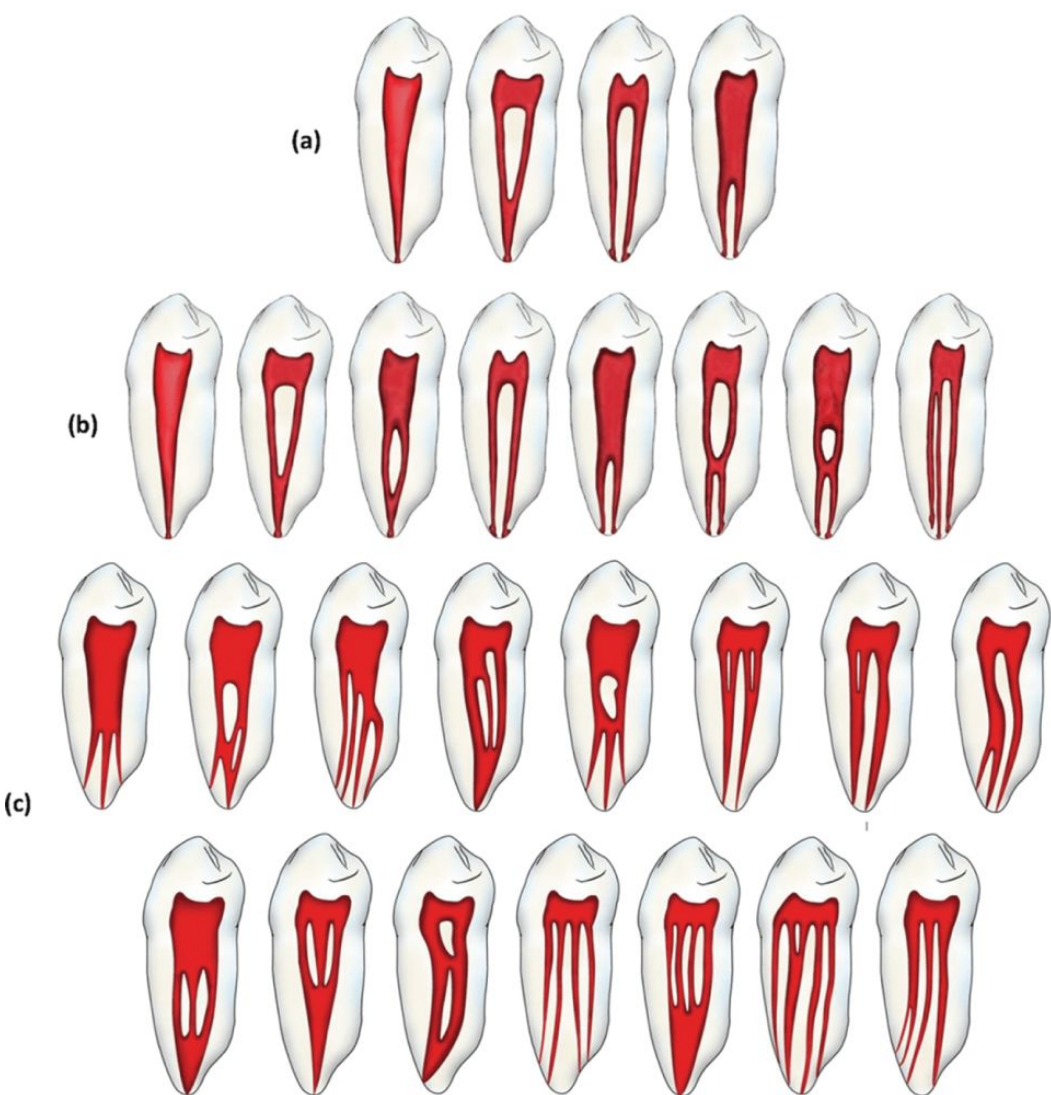
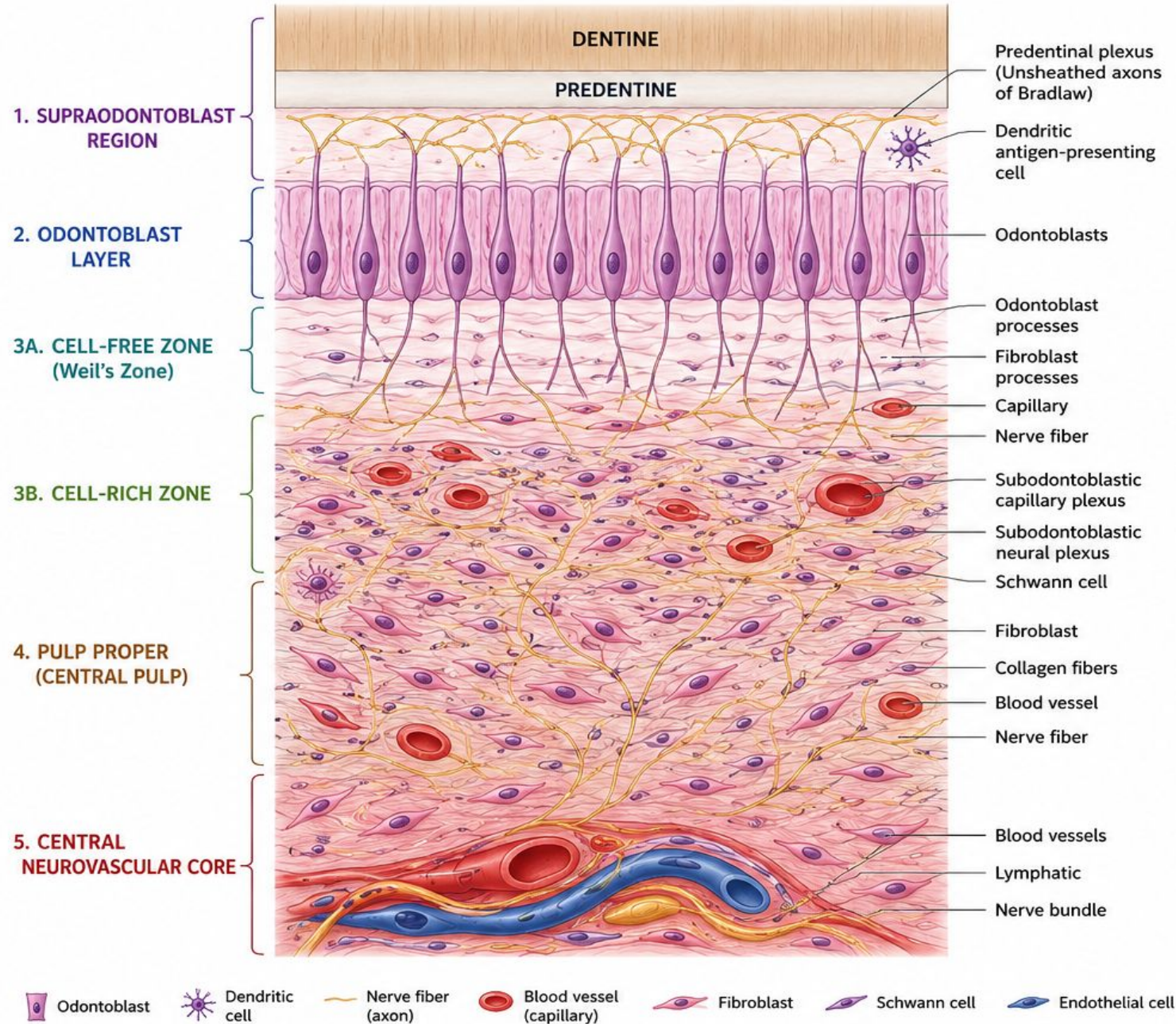


Fig. 10.2 (A) Ground section of root showing a lateral branch from the main root canal (arrow) ($\times 12$). (B) Radiograph of a root-filled tooth showing accessory canals (arrows). (A) Courtesy Dr M E Atkinson. (B) Courtesy Dr J Souyave.

REGIONAL STRUCTURE OF DENTAL PULP

(From Periphery to Core)



REGION	MAIN FEATURES
1. SUPRAODONTOBLAST REGION	- Between odontoblast layer and predentine - Unsheathed axons (predental plexus) - Dendritic antigen-presenting cells - Detects external stimuli
2. ODONTOBLAST LAYER	- Layer of odontoblasts - Responsible for dentine formation - Main effector system of the pulp
3A. CELL-FREE ZONE (Weil's Zone)	- Immediately beneath odontoblasts - Appears cell-free (actually anuclear) - Contains cell processes, capillaries, nerves - Usually in coronal pulp (after eruption) - Absent in radicular pulp - Likely an artefact of histological processing
3B. CELL-RICH ZONE	- Beneath cell-free zone - Rich in cells - Contains subodontoblastic capillary and neural plexus - Schwann and endothelial cells give "cell-rich" appearance
4. PULP PROPER (CENTRAL PULP)	- Bulk of the pulp - Loose connective tissue - Rich blood and nerve supply - Provides support, nutrition and maintenance
5. CENTRAL NEUROVASCULAR CORE	- Central core of pulp - Major blood vessels, lymphatics, nerve bundles - Prominent in root canal - Less distinct in crown due to branching

REGIONS OF PULP	LOCATION	MAIN COMPONENTS	FUNCTIONS
Supraodontoblast Region	Between odontoblast layer and predentine	• Unsheathed axons (Predentinal Plexus of Bradlaw) • Dendritic antigen-presenting cells	• First region to detect external stimuli • Senses dentinal fluid movement and extracellular fluid changes • Immune surveillance
Odontoblast layer	Peripheral layer adjacent to predentine	Odontoblasts	• Formation of primary, secondary, and tertiary dentine • Main effector system of the pulp
Cell-Free Zone (Zone of Weil)	Immediately beneath the odontoblast layer	• Odontoblast processes • Fibroblast processes • Capillaries • Nerve fibers	• Appears cell-free in routine histology but is actually anuclear • Provides pathway for nerves and vessels • Usually present in coronal pulp and absent in radicular pulp
Cell rich zone	Beneath the cell-free zone	• Numerous cells • Subodontoblastic capillary plexus • Subodontoblastic neural plexus • Schwann and endothelial cells	• Rich vascular and neural support • Contributes to pulp nutrition and sensory functions

Pulp Proper (Central Pulp)	Central portion of pulp	• Fibroblasts • Ground substance • Collagen fibers • Blood vessels and nerves	• Provides structural support • Nutrition and maintenance of pulp tissue • Similar to loose connective tissue
Central Neurovascular Core	Center of pulp, especially prominent in root canal	• Major blood vessels • Lymphatics • Nerve bundles	• Main vascular and neural supply of the pulp • Branches extensively in the coronal pulp

COMPOSITION

General Classification: Loose connective tissue.

Basic Makeup (by weight): 75% Water, 25% Organic Material

Key Structural Feature: Extracellular matrix (ECM) is more abundant with fewer cells.

- Cells are embedded in an ECM of fibres in a semifluid gel.
- The extracellular matrix (ECM) is a complex network of polysaccharides and proteins secreted by cells and closely associated with them
- Major ECM Component is Collagen, makes up 25–32% of the tissue's dry weight but only 3–5% of wet weight (low vs other loose connective tissues).

ROLE OF ECM

- **Structural Function (Minimal in Pulp):**
 - Normally acts as a scaffold to stabilise tissue.
 - However, in the dental pulp, this skeletal function is **minimal** due to the surrounding rigid hard shell (dentin/enamel).
- **Active Regulatory Function (Crucial):**
 - The ECM plays a very active role in controlling cell behaviour.
 - It controls cell development, migration, division, shape, and function

COMPOSITION IS NEVER FIXED

- ECM composition is **dynamic**, not static.
- It changes during **development, aging, injury, and exposure to bacterial toxins**.
- Different cells can **activate, migrate, differentiate**, and contribute varying ECM components over time.
- Studying ECM is challenging because:
 - **Cultured cells (in vitro)** may behave differently from those in tissues (in vivo).
 - **Animal data** may not fully represent human tissues.
 - Continuous **molecular synthesis and breakdown (turnover)** alters composition.

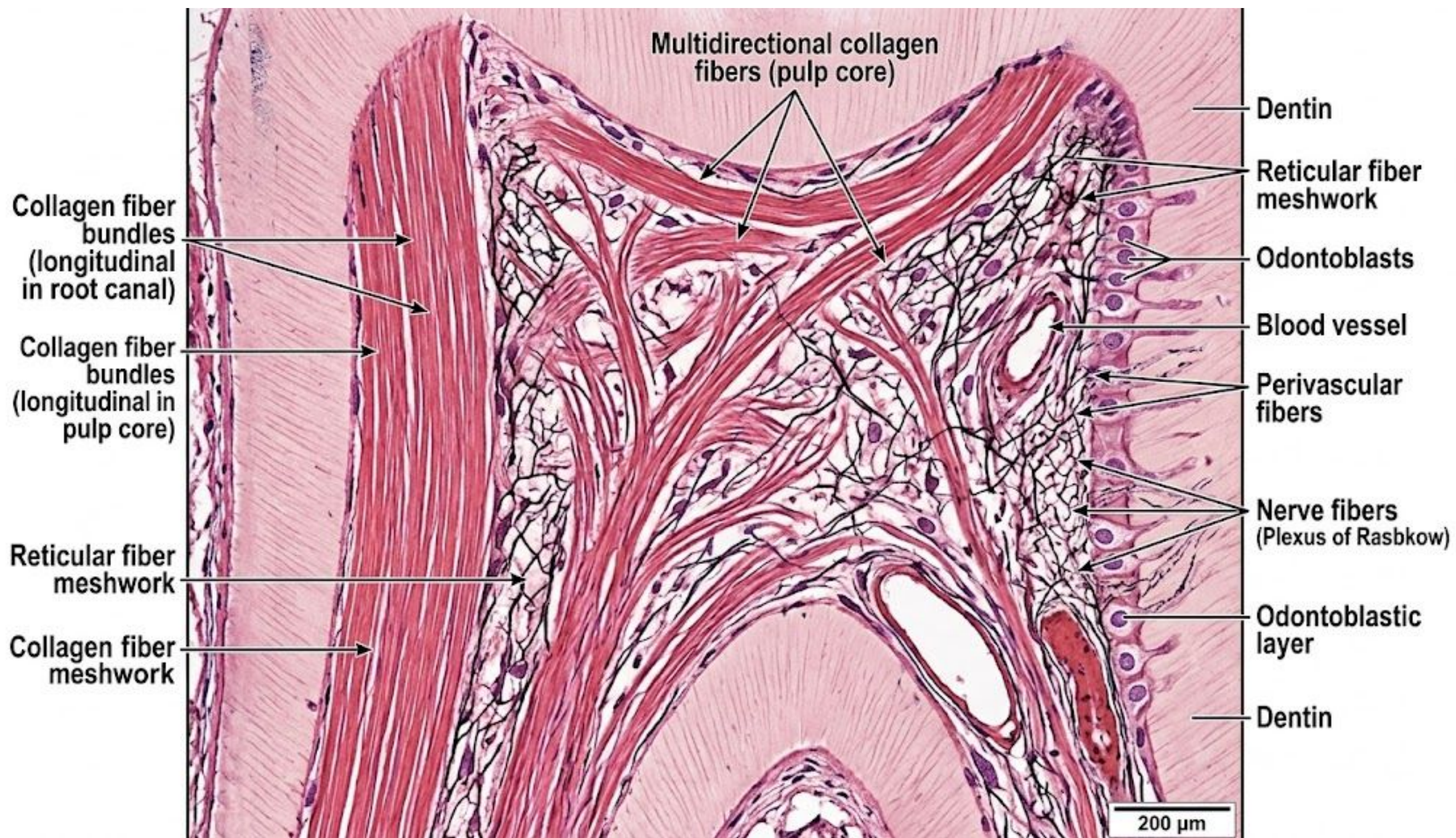
FIBRES

- **Collagen Fibers**

1. **Main fibrous component of pulp ECM**
2. **Type I collagen (60%)**
3. **Type III collagen (40%)**
4. Fibrils ~50 nm diameter; thin, irregularly scattered fibers
5. More organized peripherally, aligned parallel to the **predentin surface**

- **Other Collagen Types**

1. **Type III collagen:** only $\alpha 1$ chains; associated with tissue remodeling
2. **Small amounts of Types V & VI:** fine microfibrillar meshwork
3. **Type IV:** non-fibrous; present in blood vessel basement membranes



CONTD.

- **Non-Collagenous Fibers**

1. **Fibronectin:** A glycoprotein distributed through out the pulp, assists in cell anchorage, and maintains cell shape
 2. **Fibrillin microfibrils (10–14 nm):** beaded microfibrils, normally associated with elastic fibres elsewhere, but no convincing evidence of elastic fibres in pulp.
- **Key Point:** Dental pulp ECM is predominantly composed of **Type I and III collagen**, supported by fibronectin and fibrillin microfibrils.



Fig. 10.3 TEM showing collagen fibrils (A) within the pulp ($\times 134,000$).

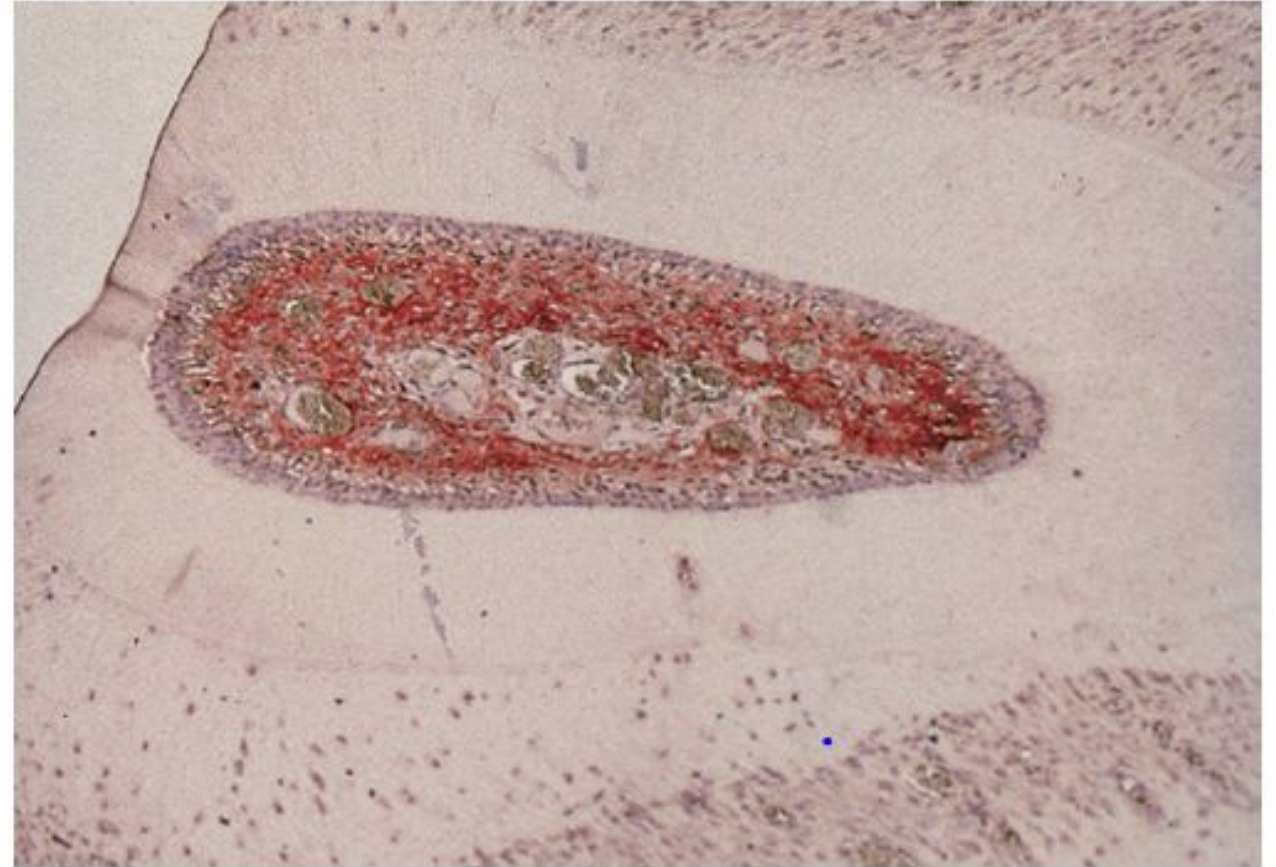


Fig. 10.4 Section of developing mouse molar root showing type III collagen stained brown by immunohistochemical methods ($\times 30$). Courtesy Dr Y Ohsaki and the editor of the *Anatomical Record*.

NON FIBROUS MATRIX

- Proteoglycans
- Glycoproteins
- Unbound glycosaminoglycans (GAGs).

GLYCOSAMINOGLYCANS (GAGs)

- Unbranched polysaccharide chains of repeating disaccharides.

4 types:

- 1. chondroitin sulphate,**
- 2. dermatan sulphate,**
- 3. heparan sulphate,**
- 4. hyaluronic acid.**

CONTD.

- Most GAGs are bound to protein core → proteoglycans (bulky, hydrophilic, form hydrated gels; reservoir for growth factors/bioactive molecules).
- Hyaluronic acid = only GAG found unbound in significant quantity; facilitates cell migration (esp. development).
- Mature pulp GAG composition: 60% hyaluronic acid, 20% dermatan sulphate, 12% chondroitin sulphate, remainder heparan sulphate.
- **Developing pulp: chondroitin sulphate is major GAG; hyaluronic acid minor.**

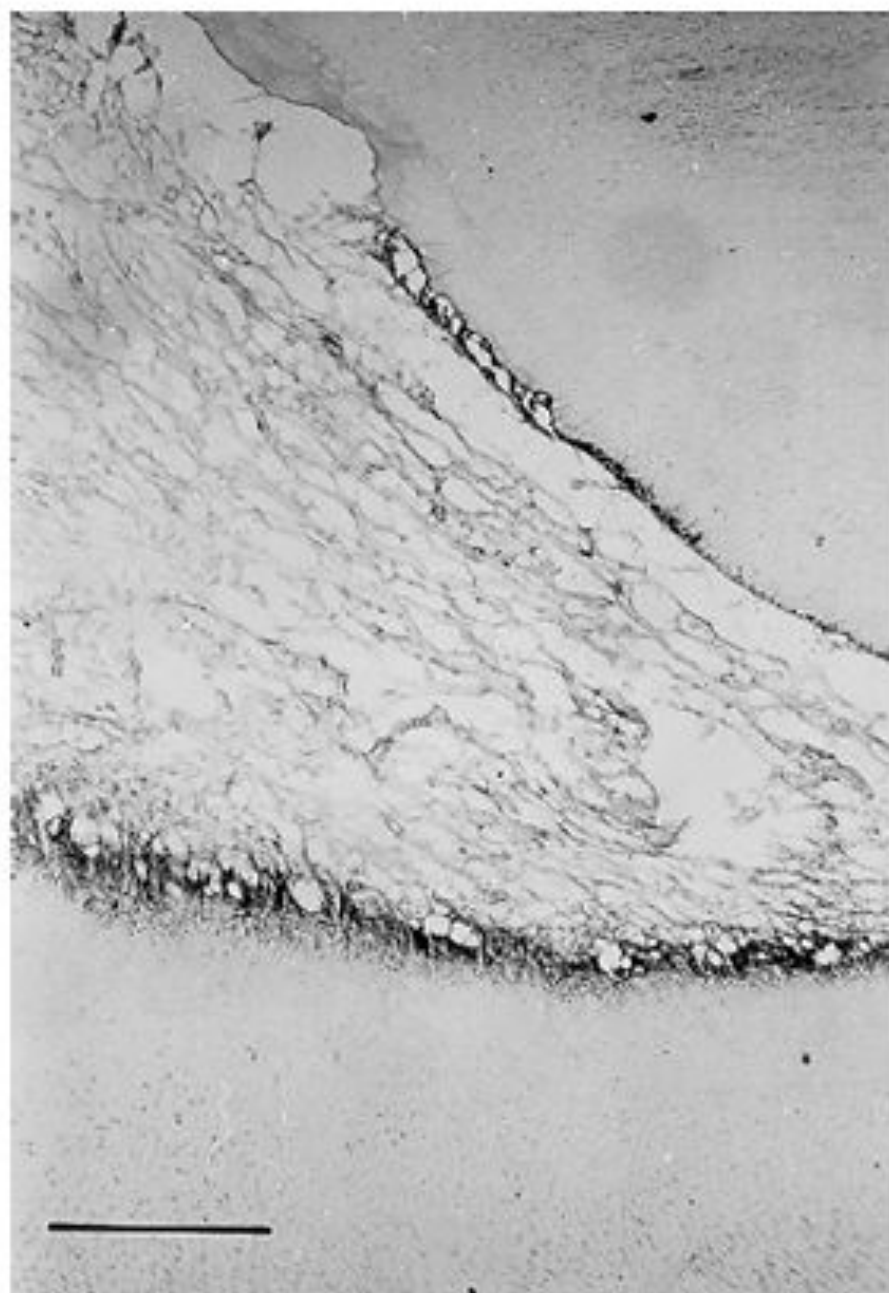


Fig. 10.5 Adult human dental pulp stained by immunohistochemical methods for chondroitin sulphate (staining black at the pulp–dentine junction), showing widespread presence but a greater concentration peripherally in the odontoblast layer, suggesting (circumstantially) a role for this molecule in the development of dentine ($\times 40$). Courtesy Dr P M Bartold, Dr U Schlagenhauf and the editor of the *International Endodontic Journal*.

PROTEOGLYCANS

- Consist of **glycosaminoglycans (GAGs)** attached to a **protein core**, functionally diverse molecules.
- **Specific Types & Locations:**
 1. **Versican:** Contributes to the **bulk of the extracellular matrix**.
 2. **Syndecan:** Found in **basement membranes** of **epithelially derived cells** (e.g., Schwann cells, endothelial cells).
 3. **Others:** Some bind to **fibres**, while others bind to **other nonfibrous tissue components**.
- **Overall Tissue Functions:**
 1. **Cementing Action:** Bind and cement various tissue components together.
 2. **Permeability Control:** Largely responsible for **limiting the tissue's permeability**, acting as a selective barrier.

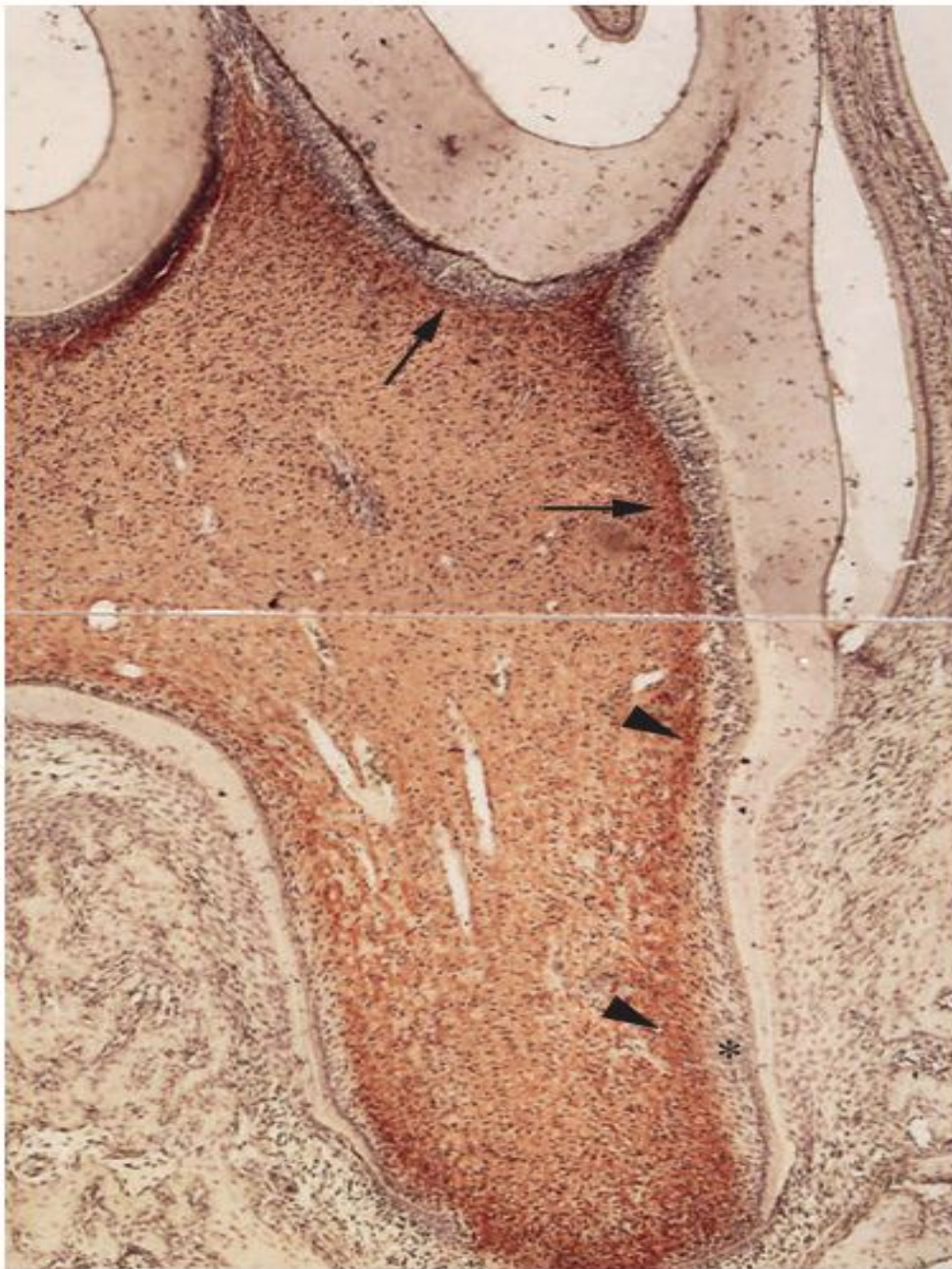


Fig. 10.6 A section of rat pulp stained immunohistochemically for the proteoglycan versican (dark stain). Staining is most intense in the subodontoblastic zone of the coronal pulp (arrows). The staining moves centrally in the cervical region (arrowheads) and is weak in the subodontoblastic zone of the root (*) ($\times 1000$). Courtesy Dr S Shibata and the editor of the *International Journal of Endodontics*.

Table 10.1 Proteoglycans of the dental pulp

Versican	Form large hydrated aggregates creating a gel
Syndecan	
Decorin	Binds to type I collagen
Biglycan	May help regulate collagen fibrillogenesis
Integrins	Cell surface adhesion receptors
Fibronectin	Concentrated near the odontoblast layer. Binds cells to extracellular matrix
Tenascin	Associated with cell movement. Concentrated near the odontoblast layer
Osteoadherin	Associated with mineralization

GLYCOPROTEIN

- Collagen itself is technically a glycoprotein.
- Fibronectin & Tenascin — highest concentration near odontoblast layer → suggested role in dentine deposition.
- Cell adhesion molecules (mostly glycoproteins) — 4 families:
 1. Immunoglobulin superfamily
 2. Selectins — guide leukocyte diapedesis in inflammation
 3. Cadherins
 4. Integrins — anchor cells to matrix

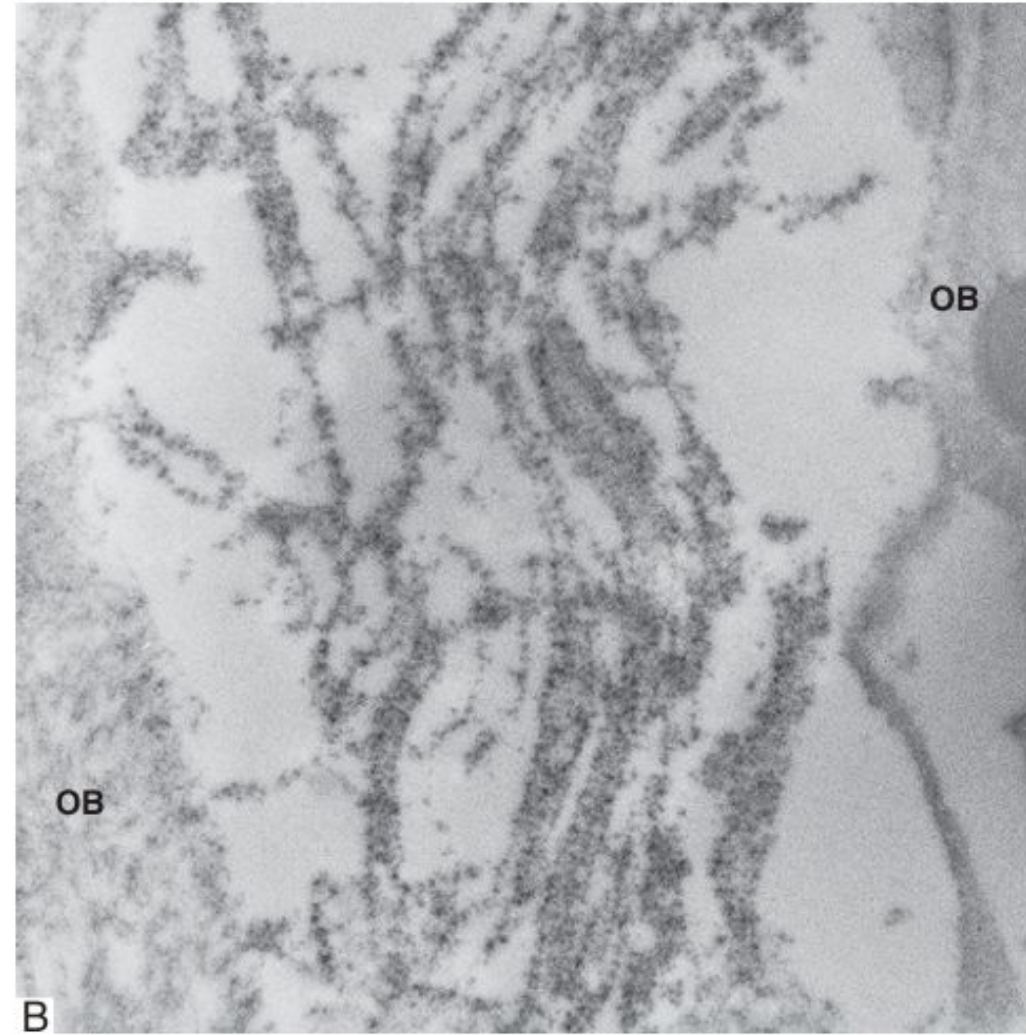
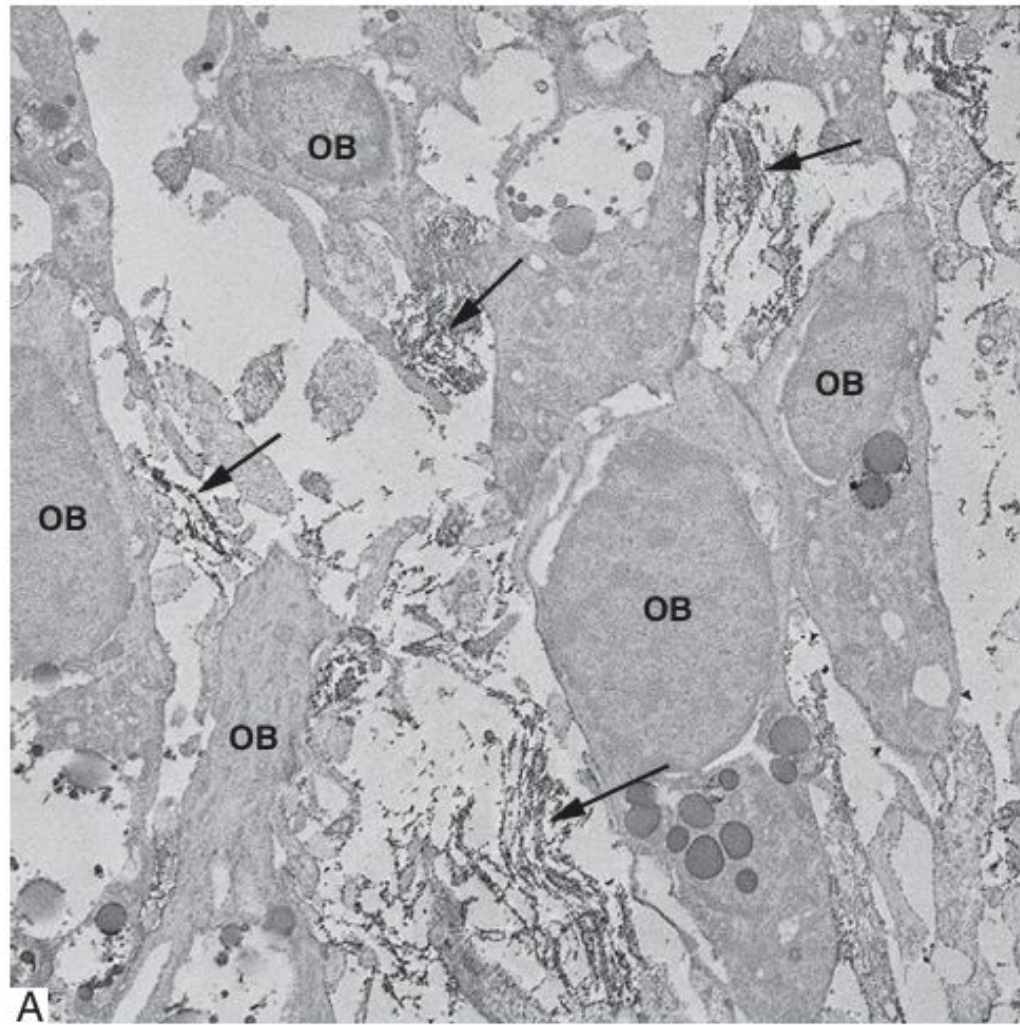


Fig. 10.7 (A) Electron micrograph showing immunolabelling for fibronectin (dark staining, arrows) on surface of collagen fibres of the odontoblast layer (OB). There is little staining in the predentine ($\times 170$). (B) Immunoelectron micrograph showing presence of fibronectin as dark-staining fibrillar fascicles between odontoblasts (OB) ($\times 3,600$). Courtesy Dr N Yoshida et al and the editor of the *Archives of Oral Biology*.

BASEMENT MEMBRANE AND DENTINE SPECIFIC PROTEINS

- **Basement Membranes**

1. Present around **epithelial cells, Schwann cells, and endothelial cells.**
2. Composed of a **meshwork of Type IV collagen.**
3. Contain embedded **adhesion and bioactive molecules.**
4. Provide structural support and regulate cell behavior.

- **Dentin-Specific Proteins**

1. **Dentin Sialoprotein (DSP) and Dentin Phosphoprotein (DPP)** are produced by **odontoblasts.**
2. Detectable in the **peripheral pulp** near the odontoblast layer.
3. Associated with **dentin formation and mineralization.**

CELLS OF DENTAL PULP

- ODONTOBLASTS
- FIBROBLASTS
- STEM CELLS
- DEFENCE CELLS

ODONTOBLASTS

- Odontoblasts are specialized cells responsible for the formation of dentine.
- They originate from ectomesenchymal cells of the dental papilla.
- In fully developed teeth, they remain viable and continue dentine formation throughout the life of the tooth

STRUCTURE OF ODONTOBLAST

- Polarized **columnar cells**
- Long **odontoblastic process** extends into dentinal tubules
- Cell body: **50 μm long, 5–10 μm wide.**
- Numerous lateral processes connect adjacent odontoblasts.
- Basally placed nucleus and abundant, **Rough Endoplasmic Reticulum (RER), Golgi Apparatus, Mitochondria**

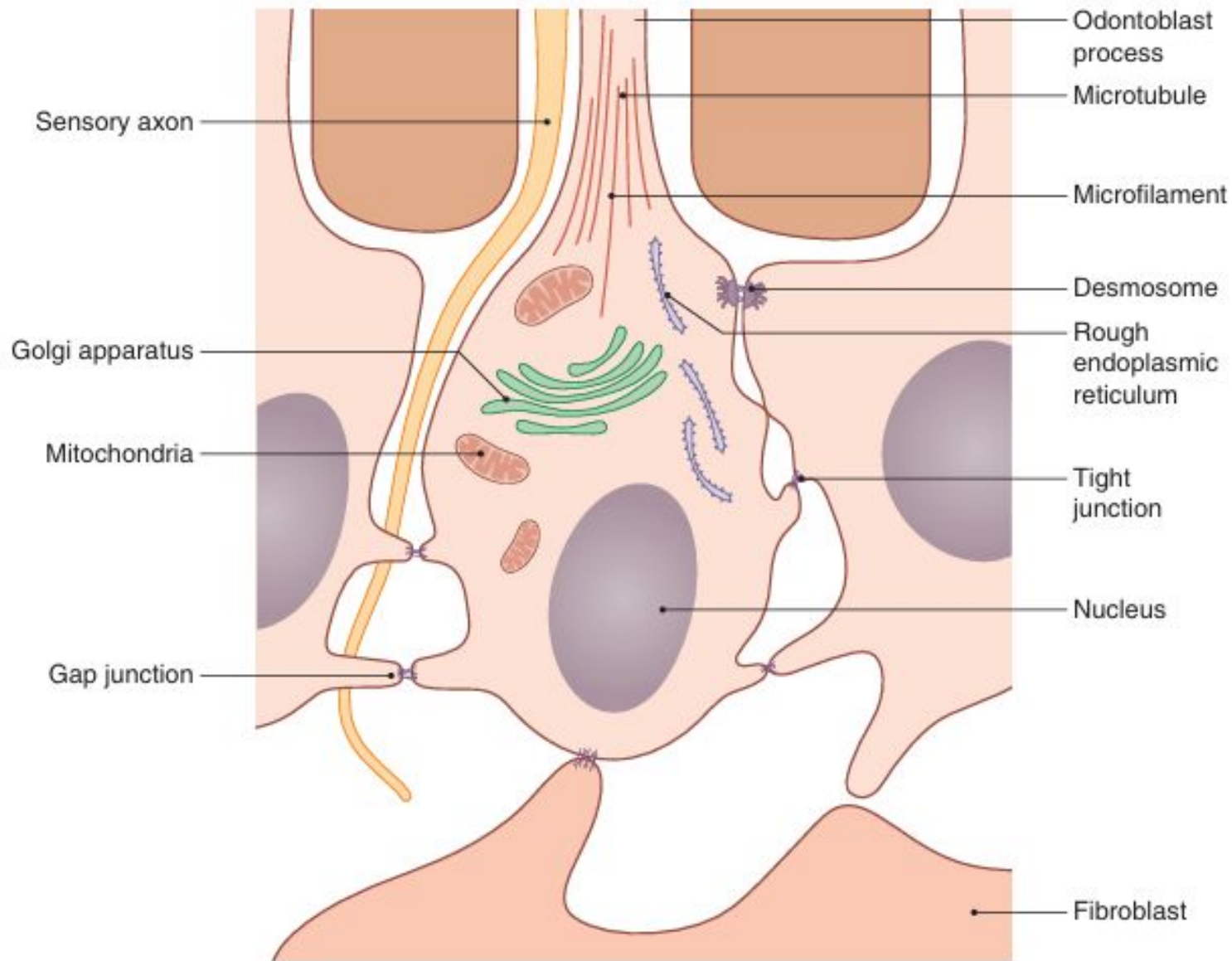
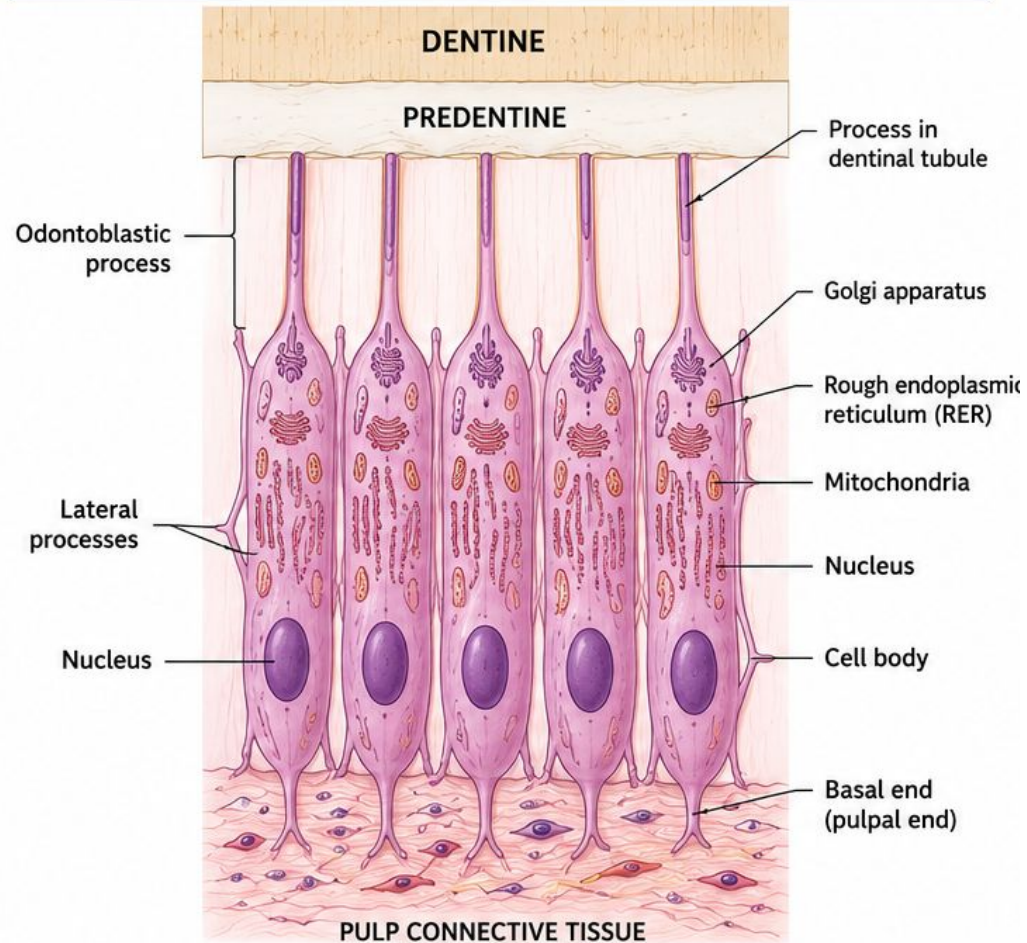


Fig. 10.12 The ultrastructure of the odontoblast in a mature tooth is typical of a polarised protein-synthesising cell of low activity containing all the organelles associated with this process (mitochondria, rough endoplasmic reticulum, Golgi apparatus) in the supranuclear region. One large process enters a dentinal tubule but many smaller ones link the odontoblast to its neighbouring odontoblasts and fibroblasts.

ODONTOBLAST – Structure



KEY FEATURES OF ODONTOBLASTS

- Polarized columnar cells.
- Cell body ~ 50 μm long and 5–10 μm wide.
- Basally placed nucleus with abundant RER, Golgi apparatus and mitochondria.
- Long odontoblastic process in dentinal tubule.
- Connected to adjacent cells by lateral processes.
- Form a single layer along predentine.

ARRANGEMENTS OF ODONTOBLASTS

Mature Tooth

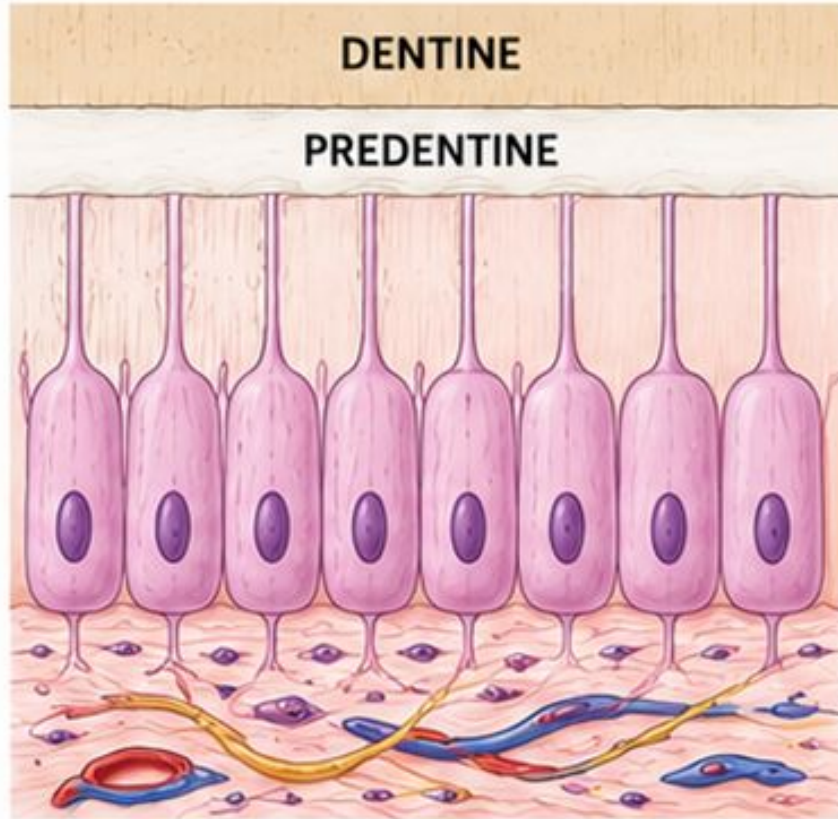
- Form a **single layer** along predentine.
- **Coronal odontoblasts:** columnar.
- **Radicular odontoblasts:** cuboidal.

Pseudostratification

- Nuclei lie at different levels.
- Gives false appearance of multiple cell layers.

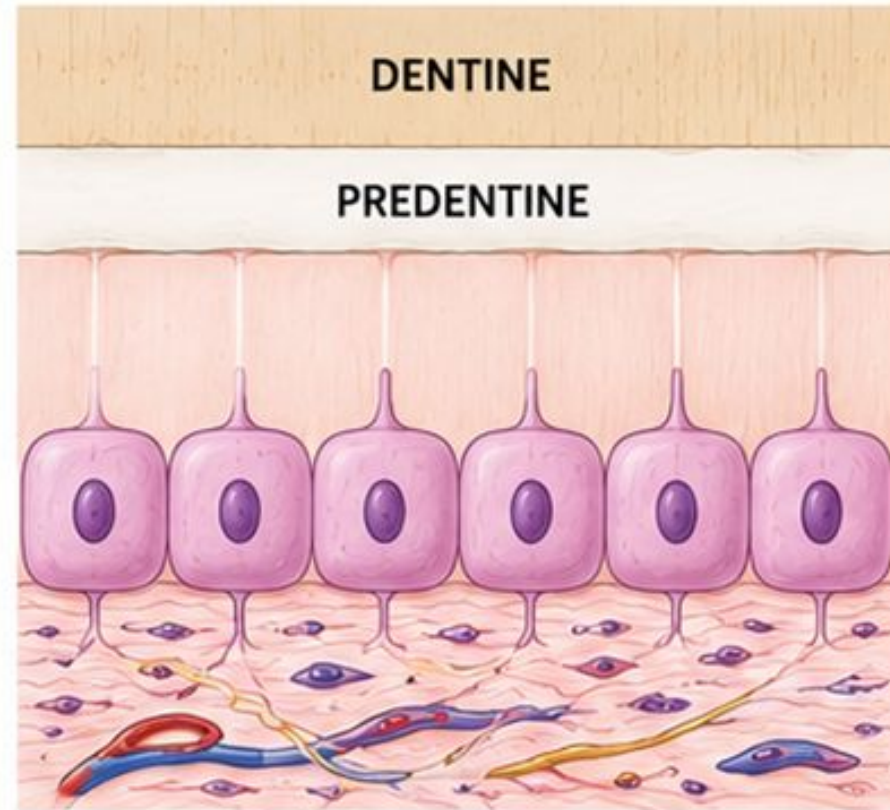
ARRANGEMENT IN MATURE TEETH

A. CORONAL PULP (Columnar Odontoblasts)



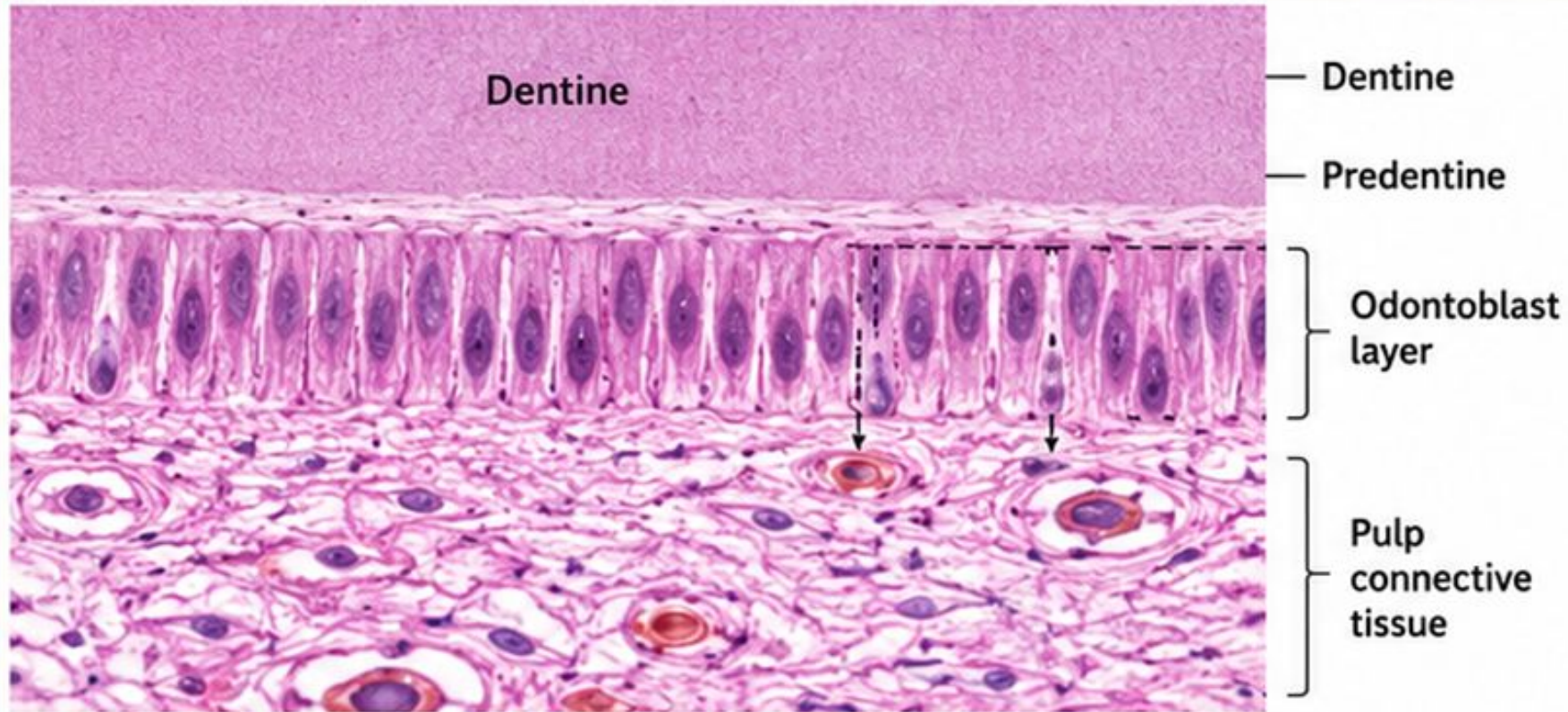
- Odontoblasts are tall and columnar.

B. RADICULAR PULP (Cuboidal Odontoblasts)



- Odontoblasts are shorter and cuboidal.

C. PSEUDOSTRATIFICATION OF ODONTOBLAST LAYER



- Nuclei of adjacent odontoblasts lie at different levels, giving the false appearance of multiple layers.

CELL JUNCTIONS IN ODONTOBLASTS

Types of Junctions

1. **Desmosomes (Macula Adherens):** Mechanical attachment.
2. **Tight Junctions:** Reduce permeability and strengthen the barrier.
3. **Gap Junctions:** Allow transfer of small molecules and coordinate odontoblast activity.

Functions

- Maintain integrity of the odontoblast layer.
- Form a selective barrier between dentine and pulp.
- Regulate permeability and cell communication.

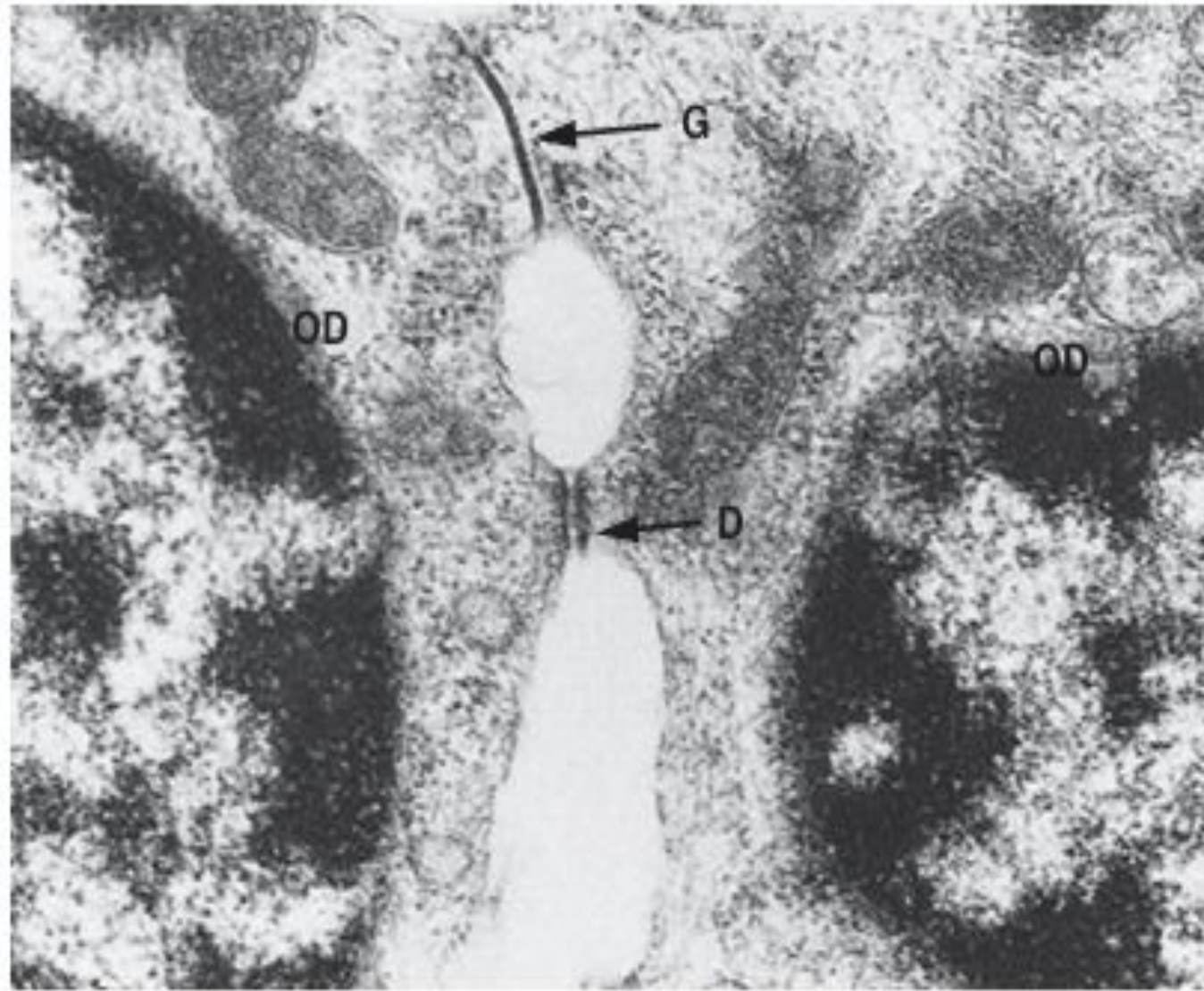
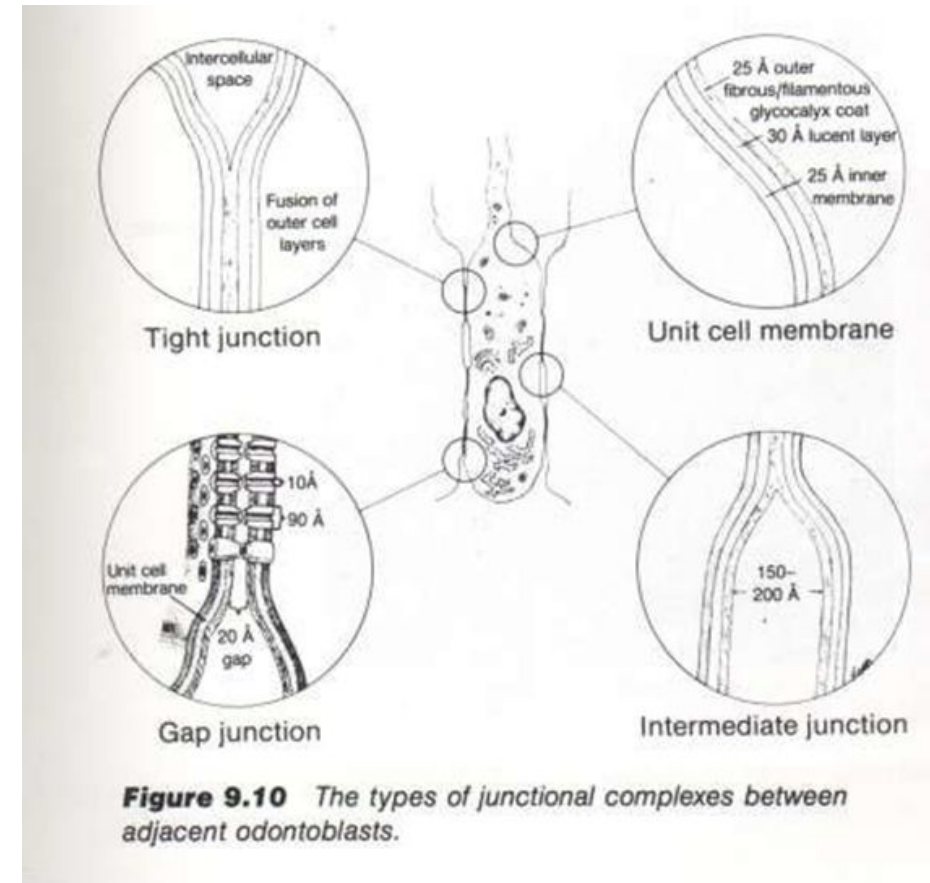


Fig. 10.14 TEM illustrating cell junctions between two odontoblast cell bodies (OD). D = desmosome; G = gap junction ($\times 45,000$).



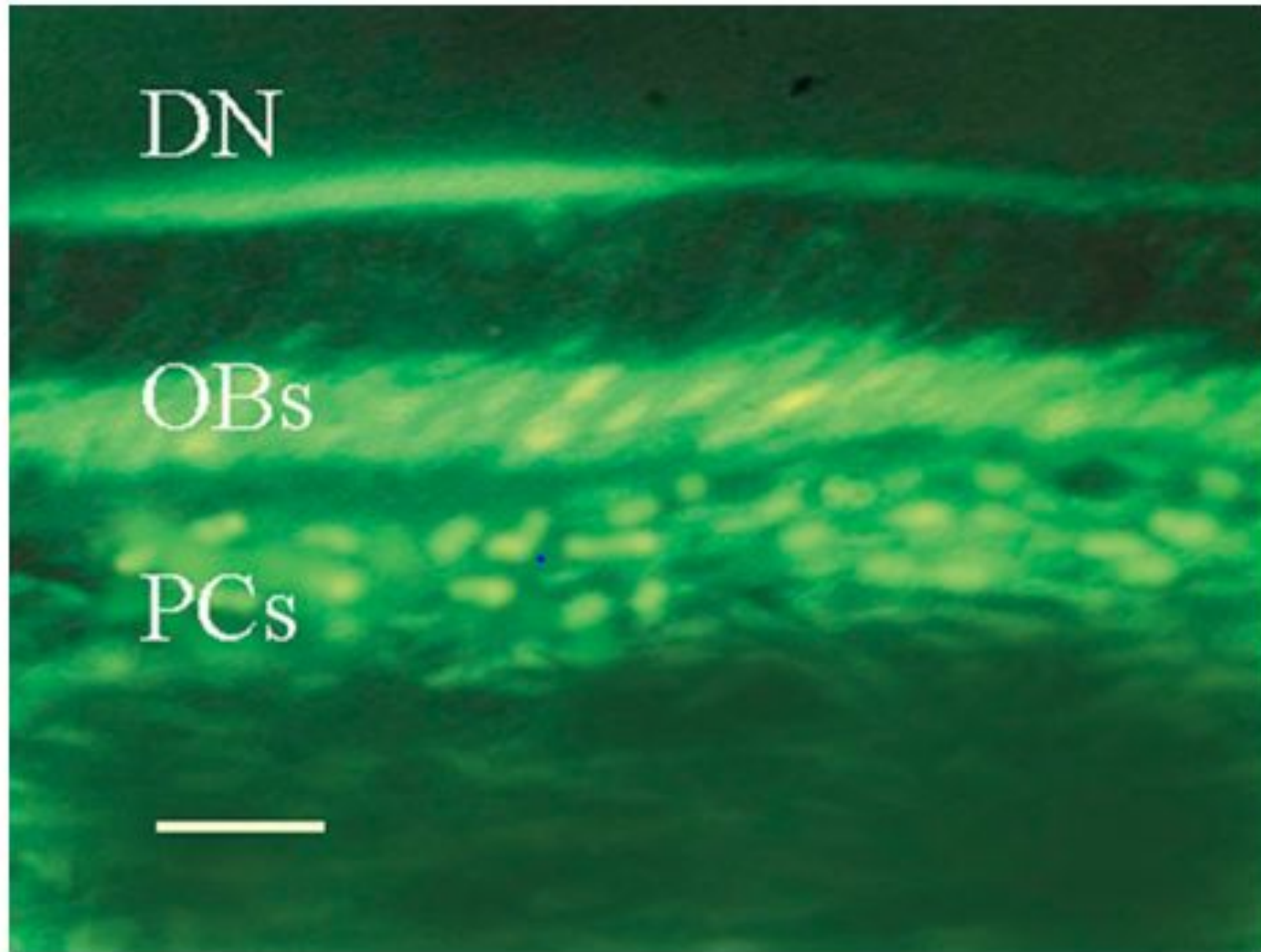
CONTD.

Communication & Signaling

- Connected with fibroblasts, stem cells, immune cells, and nerves.
- Form a functional cellular network (syncytium).
- Express growth factors: **TGF- β** and **BMPs**.

Defense & Sensory Role

- Produce **Heat shock proteins (HSPs)**, **Matrix metalloproteinases (MMPs)**, **nitric oxide**, **chemokines**, and **β -defensins**.
- Recruit immune cells and provide antibacterial defense.
- Express ion channels associated with pain signaling and dentine sensitivity.
- Possess **NGF receptors**.



Lucifer yellow dye experiment (Fig 10.15): dye applied to enamel surface passed into odontoblast layer AND underlying pulpal cells → demonstrates functional syncytium linked by junctions.

FUNCTIONS OF ODONTOBLASTS



Dentinogenic Function

- Form **primary dentine** during tooth development.
- Continuously deposit **secondary dentine** throughout life.
- Produce **tertiary (reactionary) dentine** in response to mild injury (e.g., dental caries).



Protective Function

- Act as a **selective barrier** between dentine and pulp.
- Regulate the movement of toxins and molecules reaching the pulp.
- Maintain and control the composition of **dentinal fluid**.

CONTD.

Pulp Defense Function

- Promote outward flow of dentinal fluid through dentinal tubules.
- Help dilute and remove harmful substances entering exposed dentine.
- Contribute to preservation of pulp vitality.

Immunological Function

- Recognize bacterial toxins (e.g., lipopolysaccharides).
- Produce pro-inflammatory mediators and cytokines.
- Release **Interleukin-8 (IL-8)** to recruit neutrophils during pulpal inflammation.

CONTD.

Sensory Function

- May function as sensory receptor cells.
- Contain potassium channels involved in mechanotransduction.
- Influence nearby nerve fibers by modifying the ionic environment within dentinal tubules.

FIBROBLASTS

- Most abundant cells of non-mineralized connective tissue

Structure & Organization

- Form a **loose network throughout the dental pulp**
- Connected by:
 - **Adherens junctions**
 - **Gap junctions**
- Morphology is mainly **stellate (star-shaped)**
- Cytoplasmic extensions connect:
 - Fibroblast–fibroblast
 - Fibroblast–odontoblast

CELLULAR ACTIVITY OF FIBROBLASTS

- Low turnover → relatively little RER, Golgi & mitochondria
- Make little/no contribution to dentine formation
- Can proliferate, but mitosis is rare in healthy pulp
- Apoptosis occurs → suggests normal cellular turnover

FUNCTIONS OF FIBROBLASTS

- Produce and maintain Extracellular fibres and Ground substance of pulp matrix
- Participate in Extracellular matrix synthesis
- Matrix degradation and turnover
- Minimal role in dentin formation

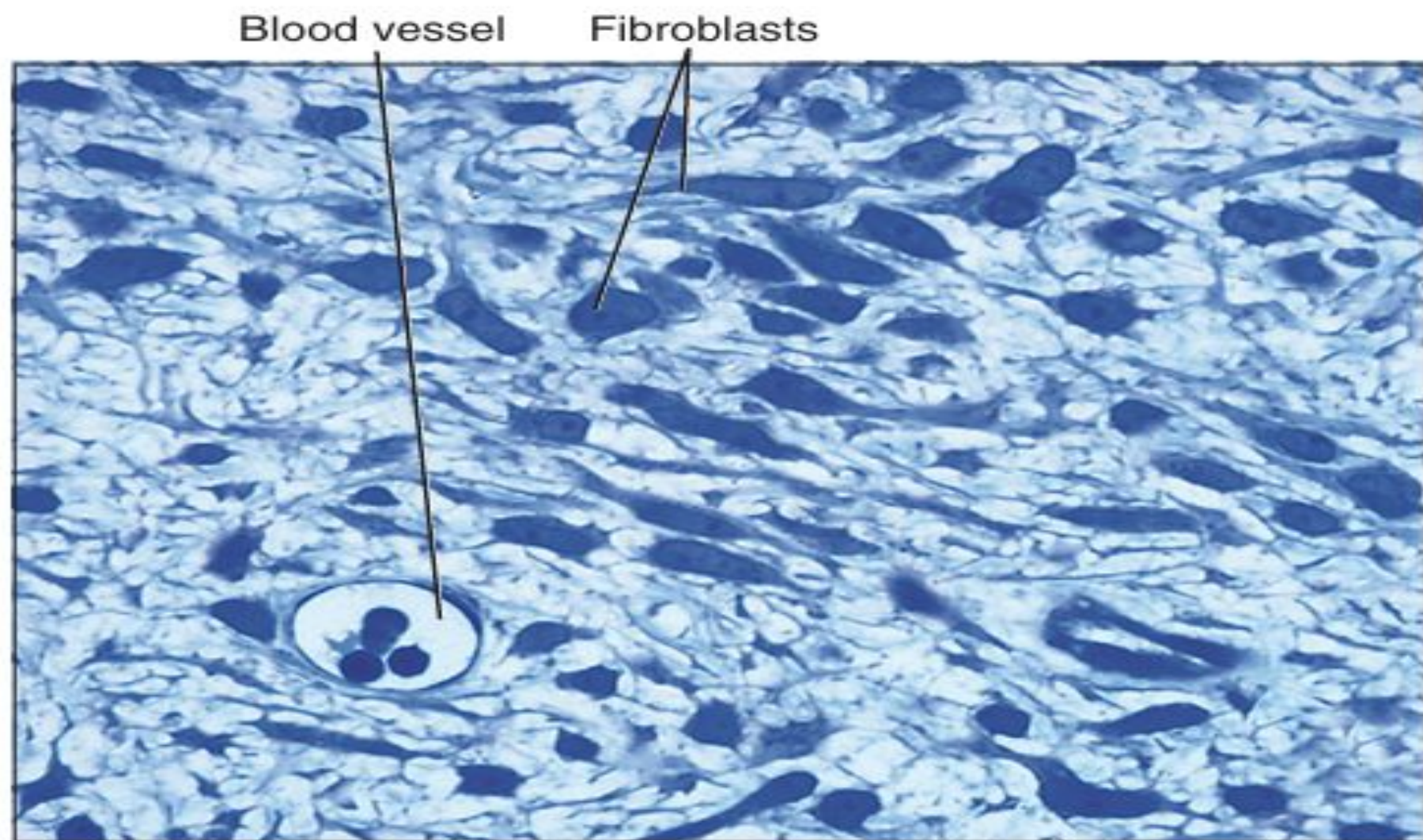
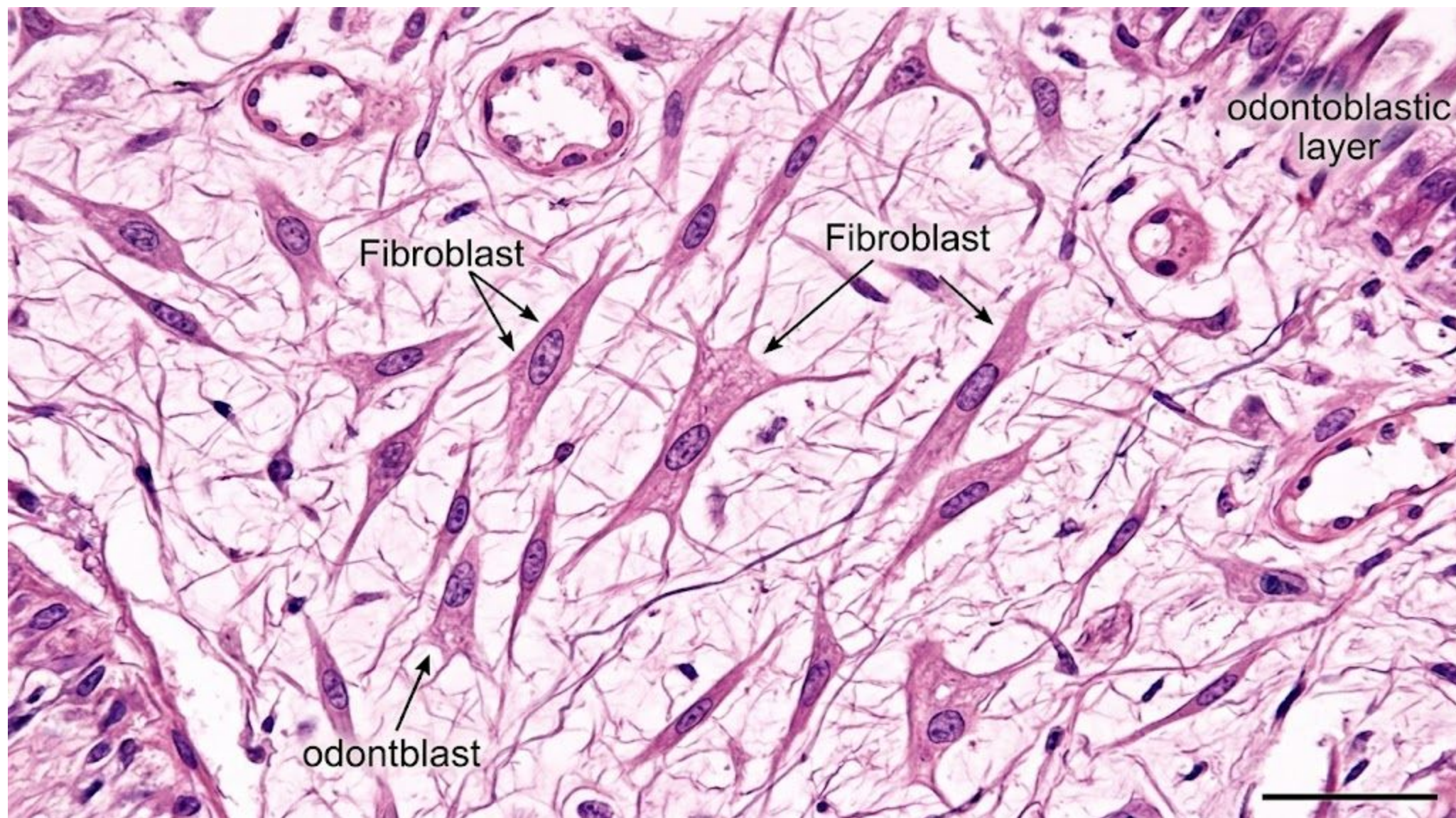
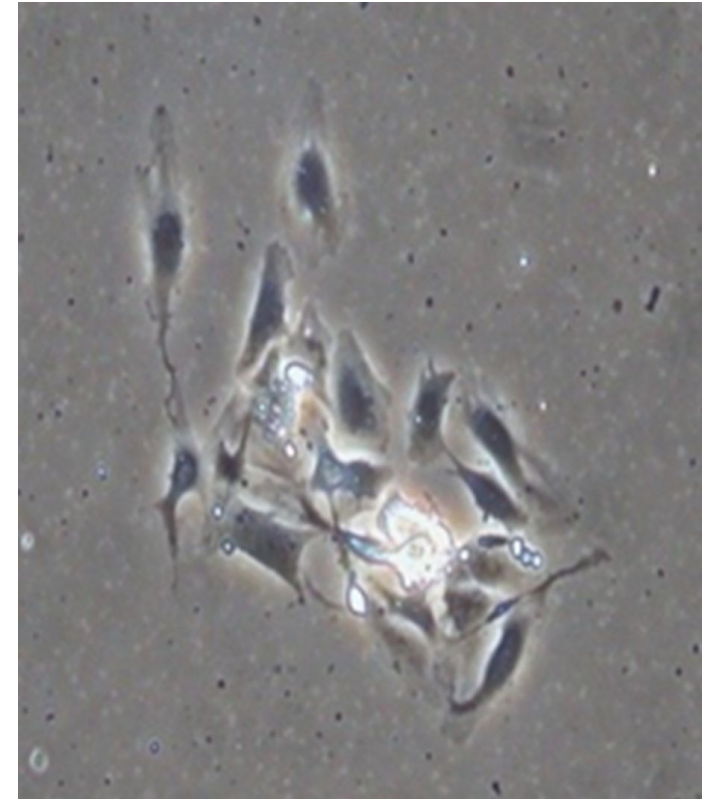


FIGURE 8-51 Light microscopic appearance of fibroblasts in the dental pulp.



STEM CELLS

- Subset of pulp cells below odontoblast layer with similar routine morphology but distinct function.
- Multipotent
- Defined by: gene expression similarity to odontoblasts + ability to differentiate and form dentine when stimulated.
- Found in 2 regions:
 1. Peripherally, beneath odontoblast layer
 2. Centrally, around blood vessels (unclear if single population)



DIFFERENTIATION POTENTIAL

DPSCs are **multipotent** and can differentiate into:

- **Odontoblast-like cells** → tertiary dentin formation
- Adipogenic cells (fat cells)
- Osteogenic cells (bone-forming cells)
- Chondrogenic cells (cartilage-forming cells)
- Neural cells
- Vascular endothelial cells

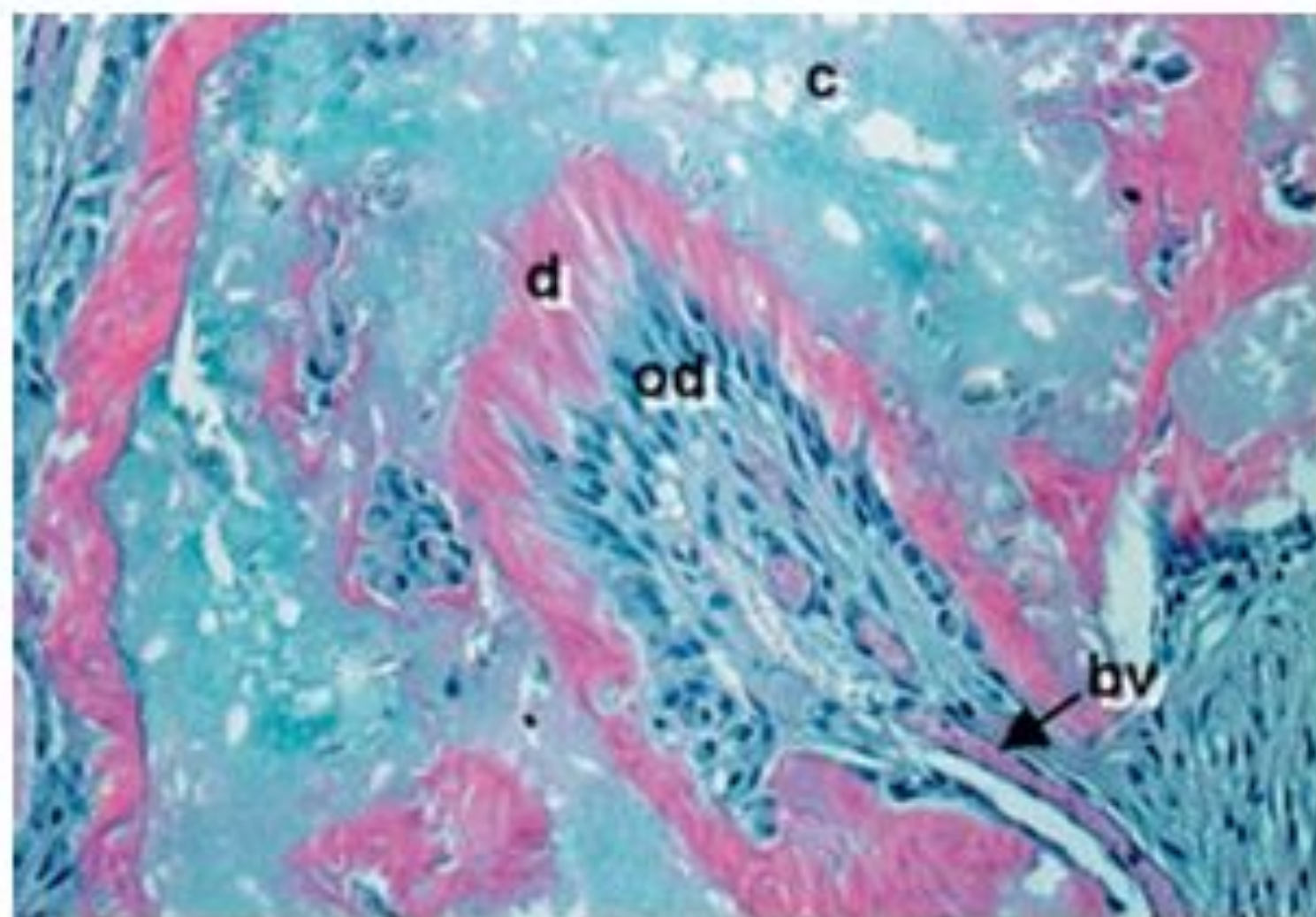


Fig. 10.17 Stem cells that were isolated from the dental pulp and then grown in culture were placed subcutaneously in mice in contact with a ceramic powder. The stem cells (od) lay down a dentine-like tissue (d) in contact with the ceramic (c). bv = blood vessel. (Polarised image.) Courtesy Dr S Gronthos et al and the editor of the *Proceedings of the National Academy of Sciences of the USA*.

CLINICAL SIGNIFICANCE

- Important for pulp regeneration and tissue repair.
- May be used for Nerve defect repair
- Regenerative therapies
- Future treatment of neurological and systemic diseases
- Signalling molecules may enhance DPSC-mediated regeneration.

DEFENSE CELLS

- T-LYMPHOCYTE
- MACROPHAGES
- DENDRITIC ANTIGEN PRESENTING CELLS
- MAST CELLS

❖ T Lymphocytes

- Present in **small numbers under normal conditions.**
- Increase significantly in response to:
 - **Pulp injury**
 - **Toxins**
 - **Inflammation**

❖ Macrophages

- Present in **substantial numbers** in dental pulp.
- Resting forms are called:
 - **Histiocytes**
 - **Pericytes**
- Difficult to differentiate from fibroblasts using routine staining.
- Identification requires **immunohistochemistry.**



Fig. 10.18 Antibody staining (black) to identify T lymphocytes in the dental pulp ($\times 500$). Courtesy Dr T Okiji and the editor of the *Journal of Dental Research*.

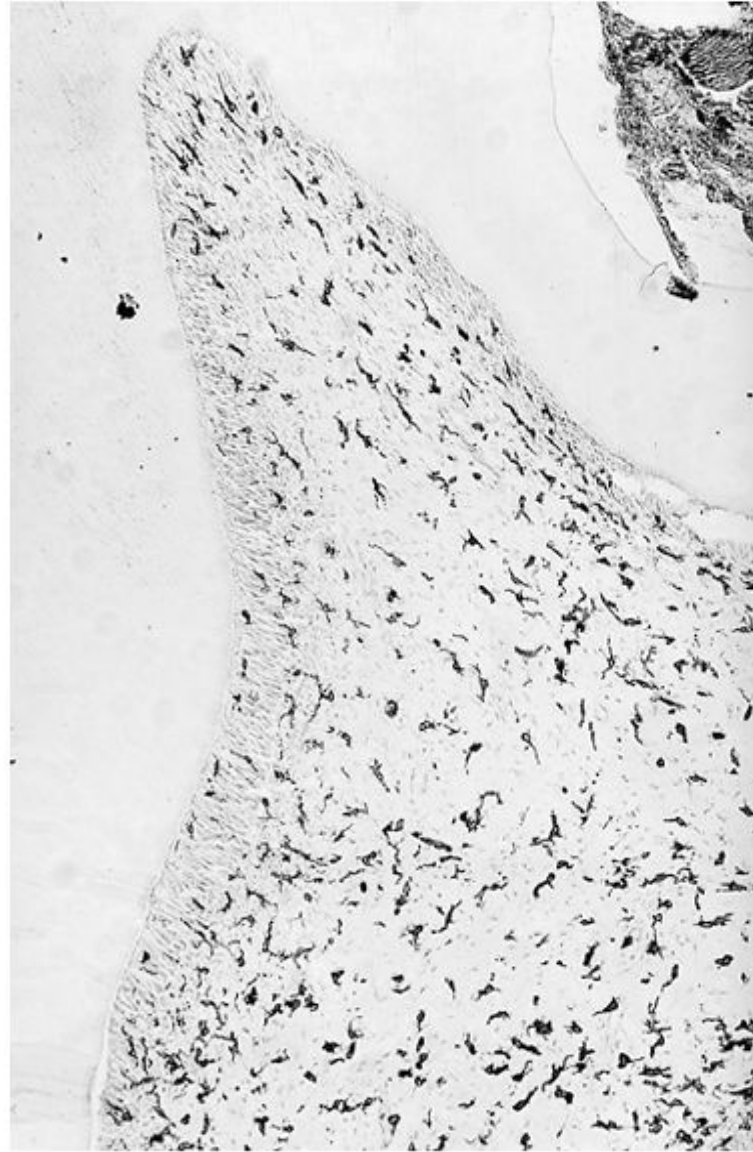


Fig. 10.19 Antibody staining (black) to demonstrate the wide distribution of macrophages in healthy rat dental pulp ($\times 50$). Courtesy Dr T Okiji and the editor of the *Journal of Dental Research*.

❖ **Dendritic Antigen-Presenting Cells**

- Large cells ($\geq 50 \mu\text{m}$) with multiple branching dendritic processes (≥ 3)
- Distribution:
 1. Throughout pulp tissue
 2. More concentrated near Odontoblast layer, nerves and blood vessels
- Some extend processes into dentinal tubules.

Functions:

1. Capture and present antigens to naïve T lymphocytes.
2. Stimulate T-cell division, activation and differentiation
3. Migrate with trapped antigens to regional lymph nodes.
4. Reach lymph nodes within 16 hours after exposure to cariogenic bacteria.
5. Mature cells express: CD273, CD86 costimulatory molecules and MHC Class II

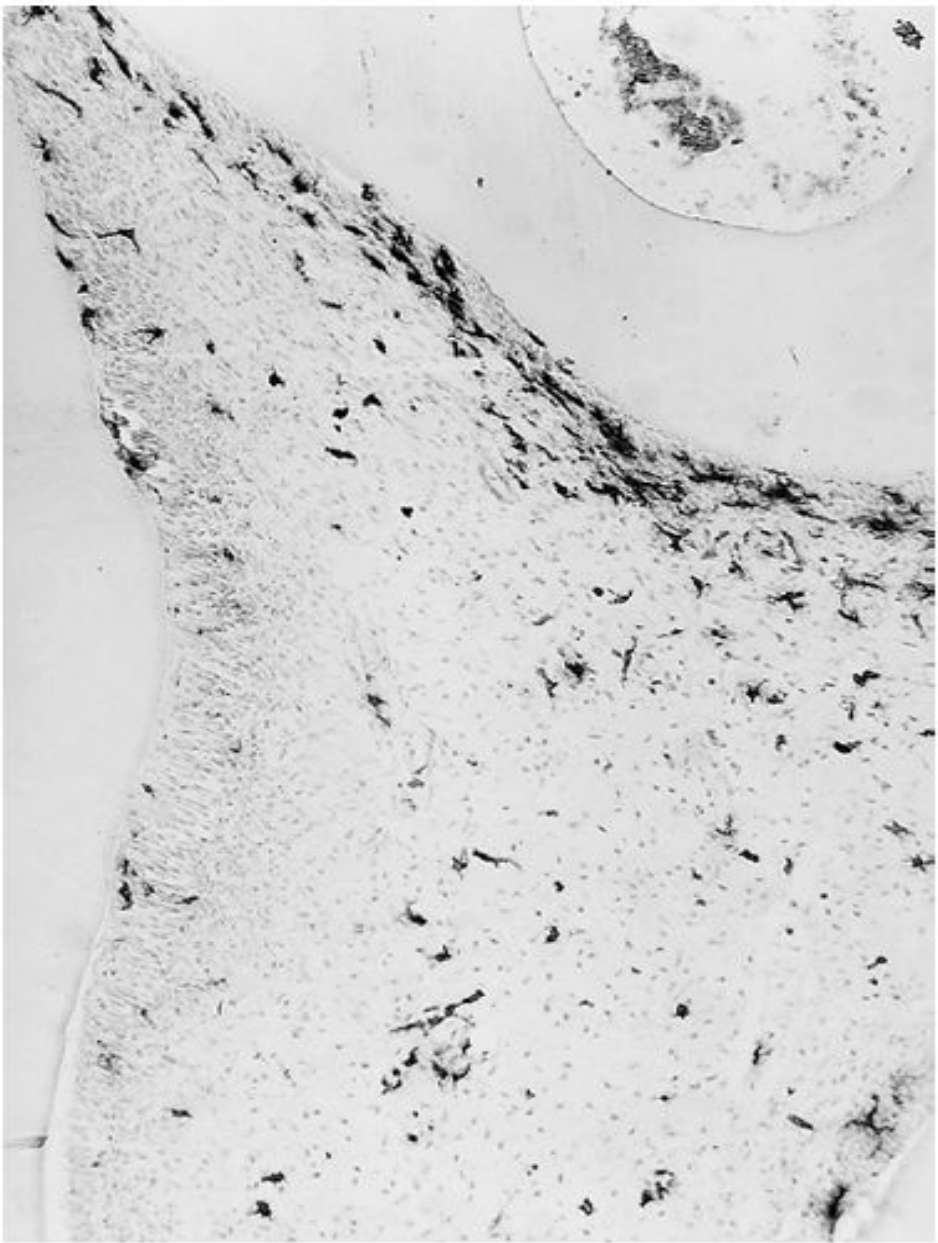


Fig. 10.20 Immunohistochemical stain (black) identifying antigen-presenting cells particularly prominent at the periphery of rat molar pulp ($\times 70$). Courtesy Dr T Okiji and the editor of *the Journal of Dental Research*.

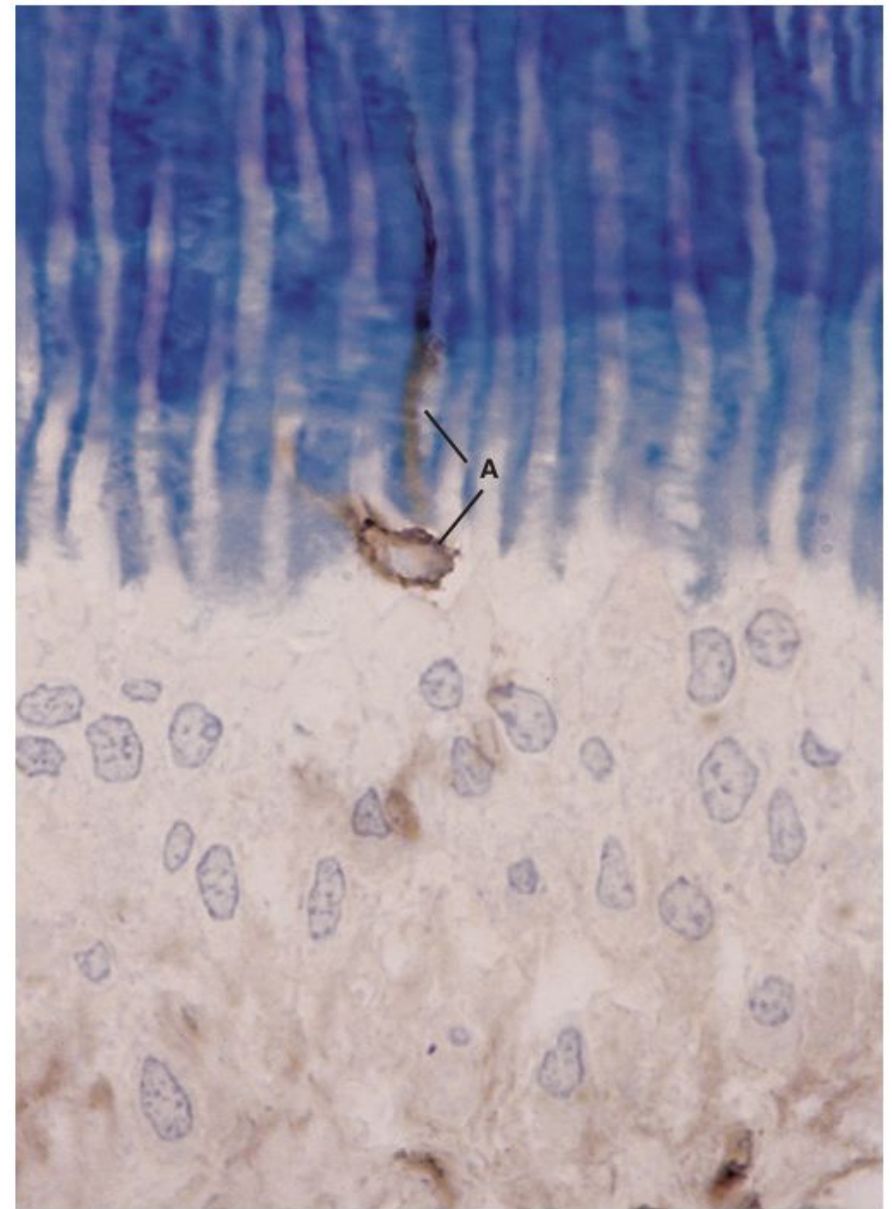


Fig. 10.22 Immunohistochemical stain (brown) identifying an antigen-presenting cell (A) in the outer odontoblast layer and extending a process into the inner dentine (methylene blue; $\times 500$). Courtesy Dr N Yoshida and the editor of *the Journal of Dental Research*.

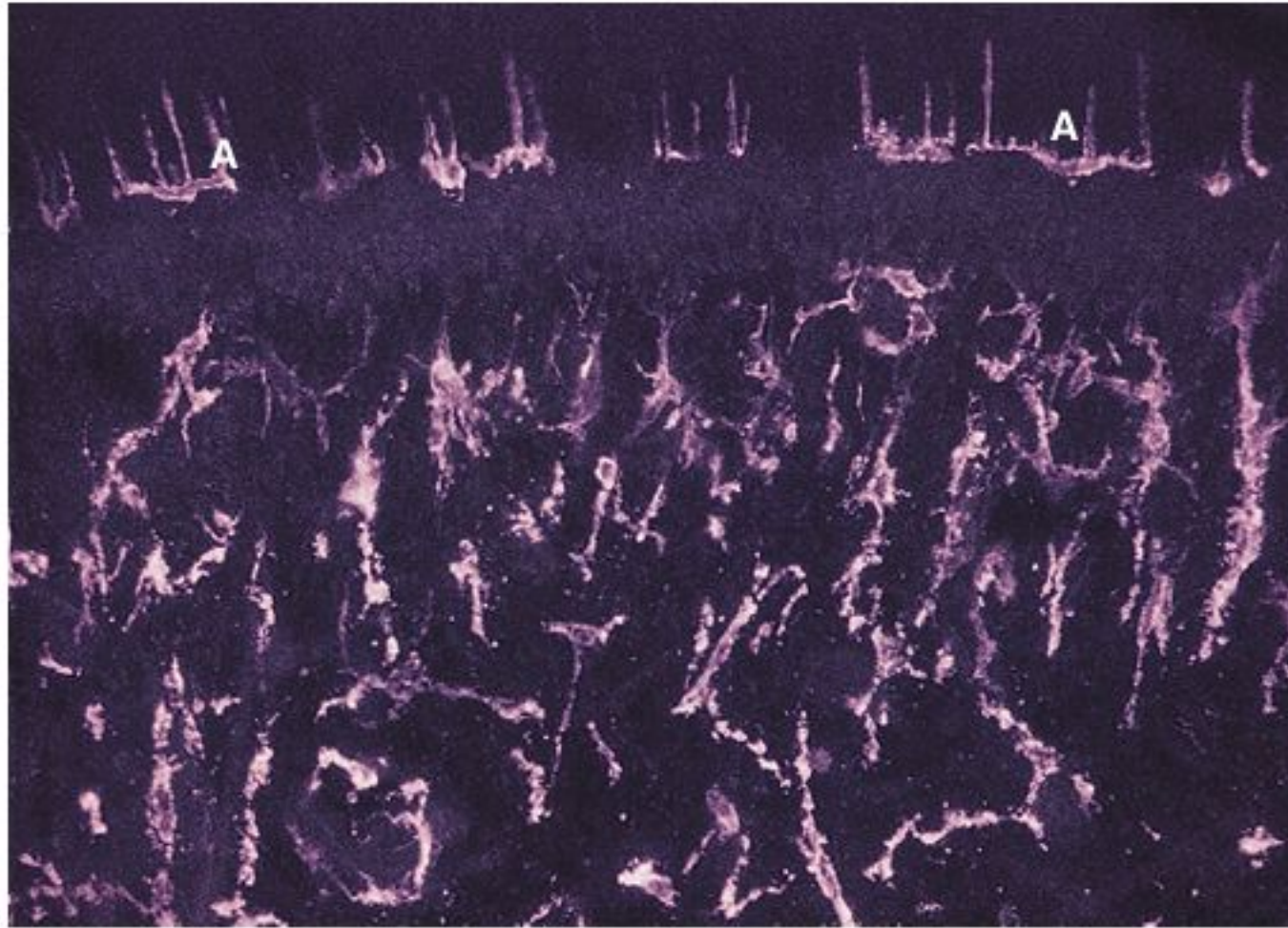


Fig. 10.21 Immunohistochemical stain (white) viewed under fluorescent light showing the distribution of antigen-presenting cells. Note the cells (A) in the periphery of the odontoblast layer have processes extending into the dentine ($\times 200$). Courtesy Dr N Yoshida and the editor of the *Journal of Dental Research*.

❖ Mast Cells

- Difficult to identify histologically.
- Fragile cells → often destroyed during tissue processing.

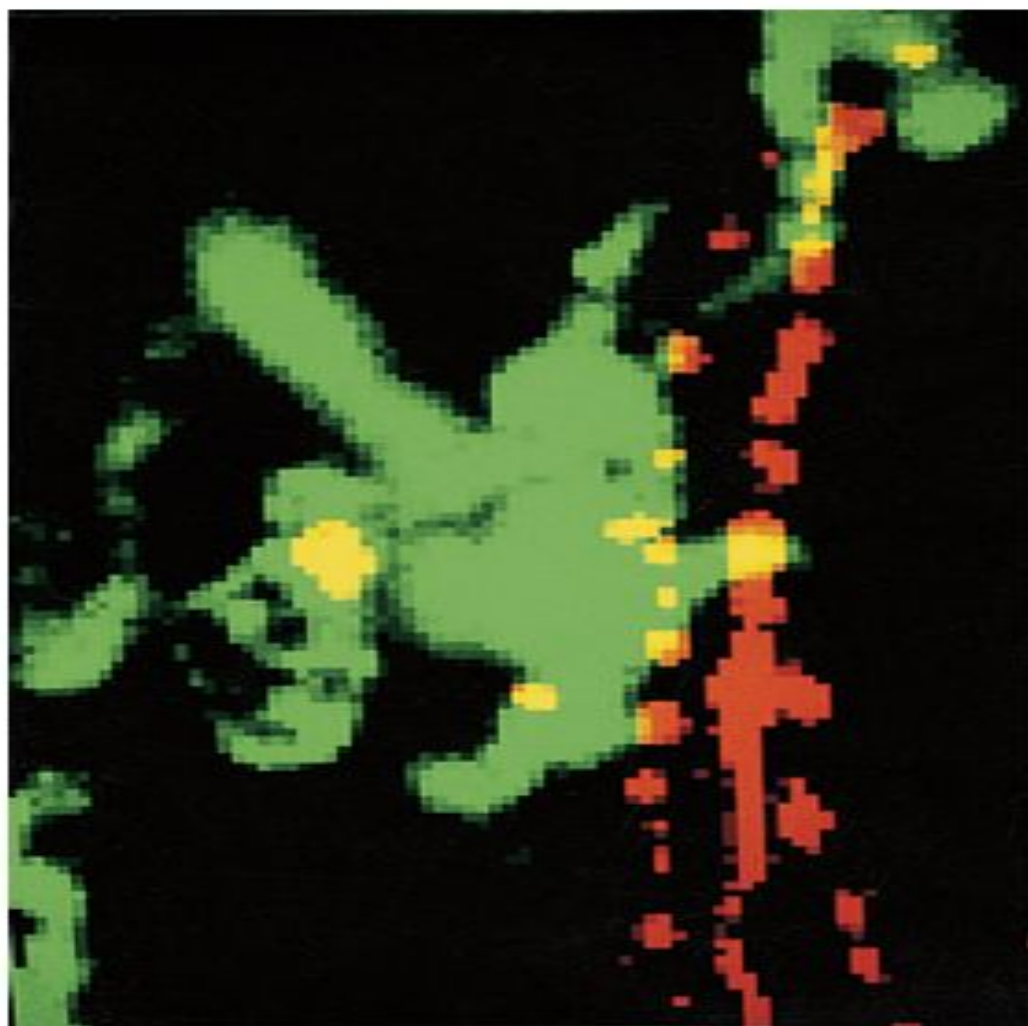


Fig. 10.23 Confocal laser scanning micrograph of a section of dental pulp from the pulp horn of a rat. It is double stained with two fluorescent dyes staining macrophages (green) and substance-P-containing nerve fibres (red). This technique allows great depth of focus and, in this case, shows areas (in yellow) where the nerve and macrophage are in close contact ($\times 1,000$). Courtesy Dr T Okiji and the editor of the *Journal of Dental Research*.

BLOOD VESSELS OF DENTAL PULP

Vascular Architecture

- Blood vessel distribution **parallels nerve distribution.**
- Arterioles and venules enter through:
 - **Apical foramina**
 - **Lateral canals**
- Enter as **neurovascular bundles.**
- Largest arterioles: approximately **150 μm diameter.**
- Vessels run longitudinally in the root canal and branch extensively in the **coronal pulp**

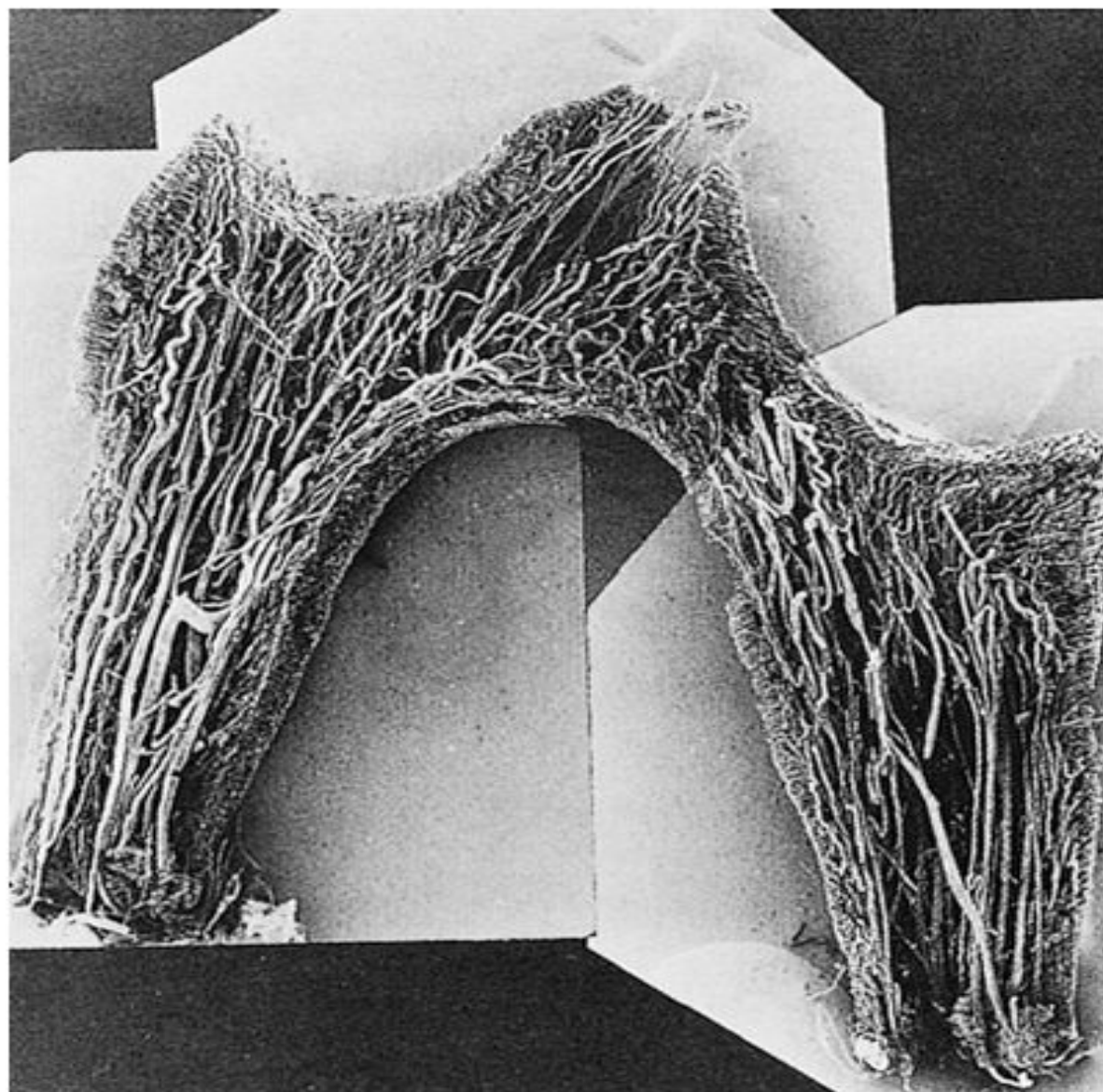


Fig. 10.26 SEM of a vascular cast of a dog mandibular molar. Arterioles run coronally along the sides of the root canal; venules drain in the centre. Larger vessels follow relatively straight pathways. Side branches subdivide to form a capillary network beneath the dentine ($\times 8$). Courtesy Dr Y Kishi, Dr K Takahashi, Dr K. Kanagawa and the editor of the *Journal of Dental Research*.

Capillary Network

- Capillary loops extend toward dentin.

Subodontoblastic capillary plexus:

- Located beneath odontoblasts.
- Capillary diameter: **6–8 μm .**

Capillaries are present:

- Within odontoblast layer
- Beneath odontoblasts
- Between odontoblasts and predentin
- **Do not enter dentinal tubules.**

Special Features

- Dentinal fluid is likely an **ultrafiltrate of pulpal interstitial fluid.**
- **4–5% capillaries are fenestrated:**
 - Fenestrations: **60–80 nm**
 - Allow rapid exchange of materials.
- Arteriovenous and venous-venous anastomoses regulate rapid perfusion changes.

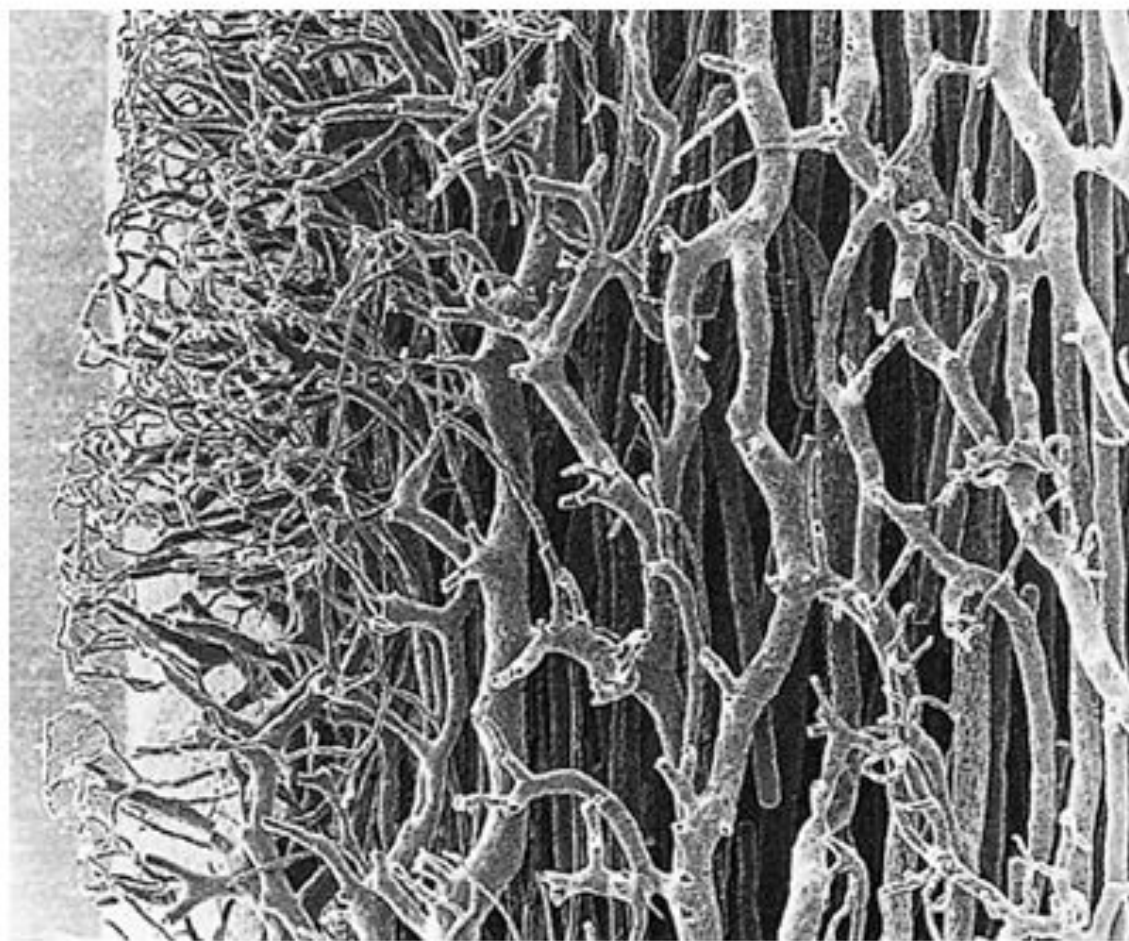


Fig. 10.27 SEM of a vascular cast of terminal capillary network in pulp. There are three layers of the terminal supply to the odontoblast layers: most superficially the flat-ended capillary loops, beneath these the capillary network and under these the venular network. Deep to all these layers are straight arterioles ($\times 11$). Courtesy Dr K Takahashi, Dr Y Kishi, Dr S Kim and the editor of the *Journal of Endodontics*.

LYMPHATICS AND VASCULAR REGULARIZATION

Lymphatic Vessels are difficult to identify in dental pulp.

- Some vessels show lymphatic features like
 1. Incomplete endothelial junctions
 2. Poor basement membrane
- Confirmed by tracing particles to regional lymph nodes.

Control of Pulpal Blood Flow

Nervous control is dominant.

- **Sympathetic nerves:**
 1. Act on arteriolar smooth muscle.
 2. Maintain vasoconstrictor tone through: **Noradrenaline and Neuropeptide Y**
- **Parasympathetic supply:** Little evidence available.

CONTD.

Sensory nerves:

- Cause vasodilation and increased permeability during inflammation.
- Main mediators:
 1. **CGRP**
 2. **Substance P**
 3. Nitric oxide

Pulpal Blood Flow

- Approximately **20–60 mL/min/100 g tissue**.
- High pulsatile interstitial pressure:
 - Drives outward dentinal fluid movement.
 - Helps wash out/dilute bacterial toxins after loss of enamel or cementum.

NERVES OF DENTAL PULP

- Dental pulp is **highly innervated**.
- Around **2500 axons** enter the apical foramen of a mature premolar.

Types of Nerve Fibres

- **Myelinated afferent fibres (25%)**
- Cell bodies in **trigeminal ganglion**.
- Mainly:
 - **A δ fibres (90%)**
 - Diameter: 1–6 μm
 - Sharp pain transmission
 - **A β fibres:**
 - Diameter: 6–12 μm
- **Unmyelinated C fibres**
- Function as:
 - Sensory fibres
 - Autonomic fibres

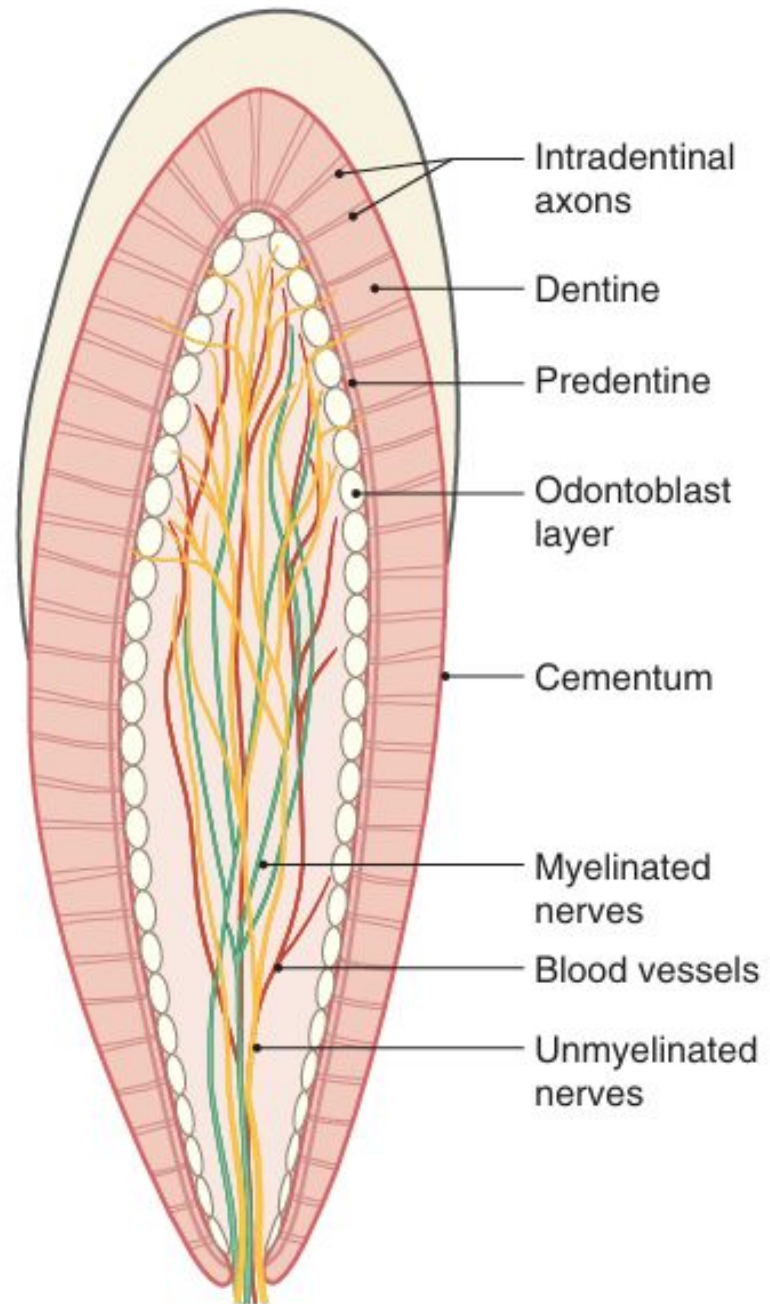


Fig. 10.38 General distribution of myelinated nerves (green), nonmyelinated nerves (yellow) and blood vessels (red) in the pulp.

CONTD.

Distribution

- Nerve bundles accompany vessels in the root.
- Branch extensively in coronal pulp.
- **Plexus of Raschkow:**
 - Dense subodontoblastic nerve plexus.
 - Develops after tooth eruption.
 - Nerves enter the tooth through the apical foramen as myelinated nerve bundles. They branch to form the subodontoblastic nerve plexus of Raschkow which is separated from the odontoblasts by a cell-free zone of Weil.
 - Branches → marginal plexus (between odontoblast layer & predentine); others continue into dentine with odontoblast processes in dentinal tubules.
 - Subodontoblastic plexus may be a key site of sensory activation.

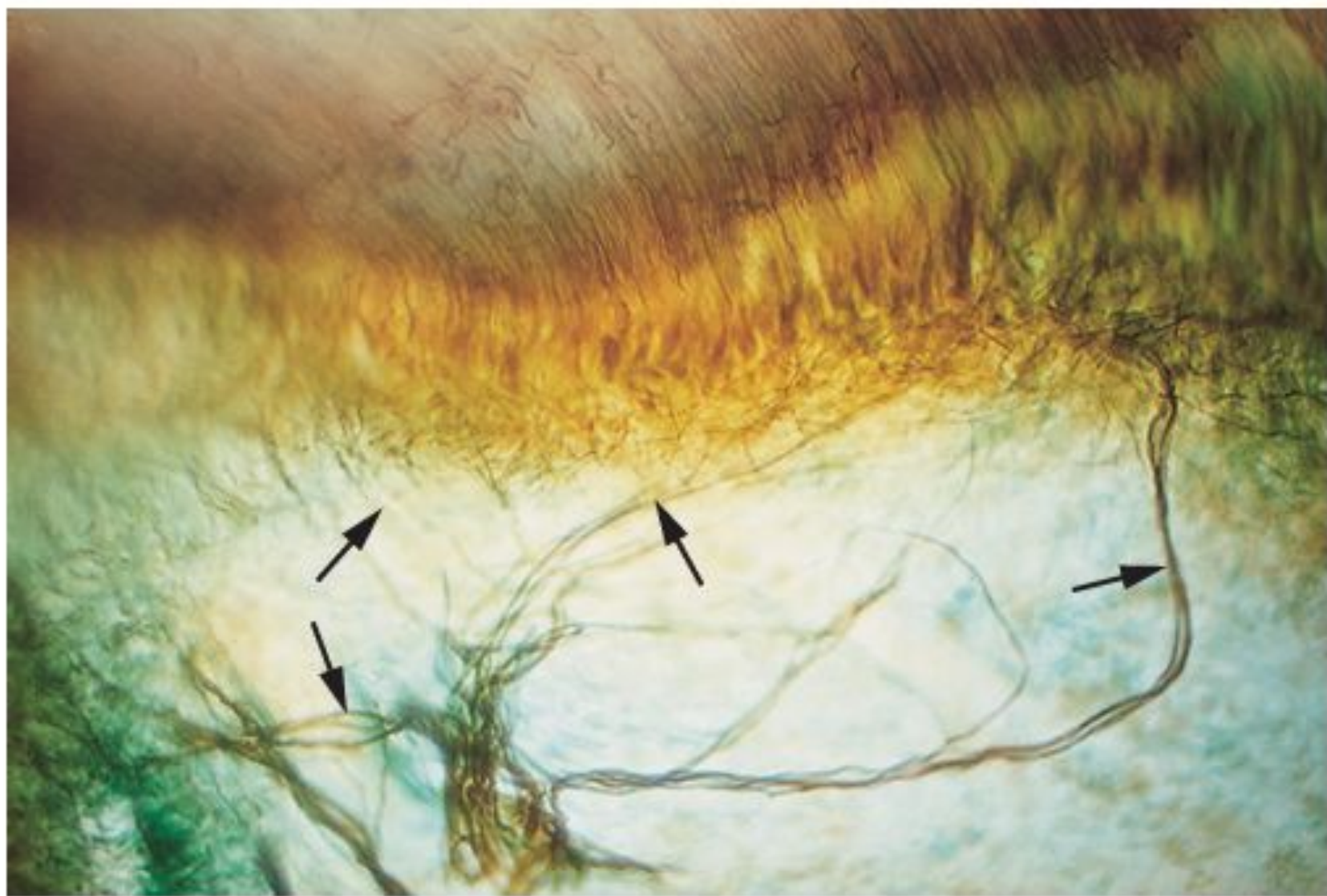
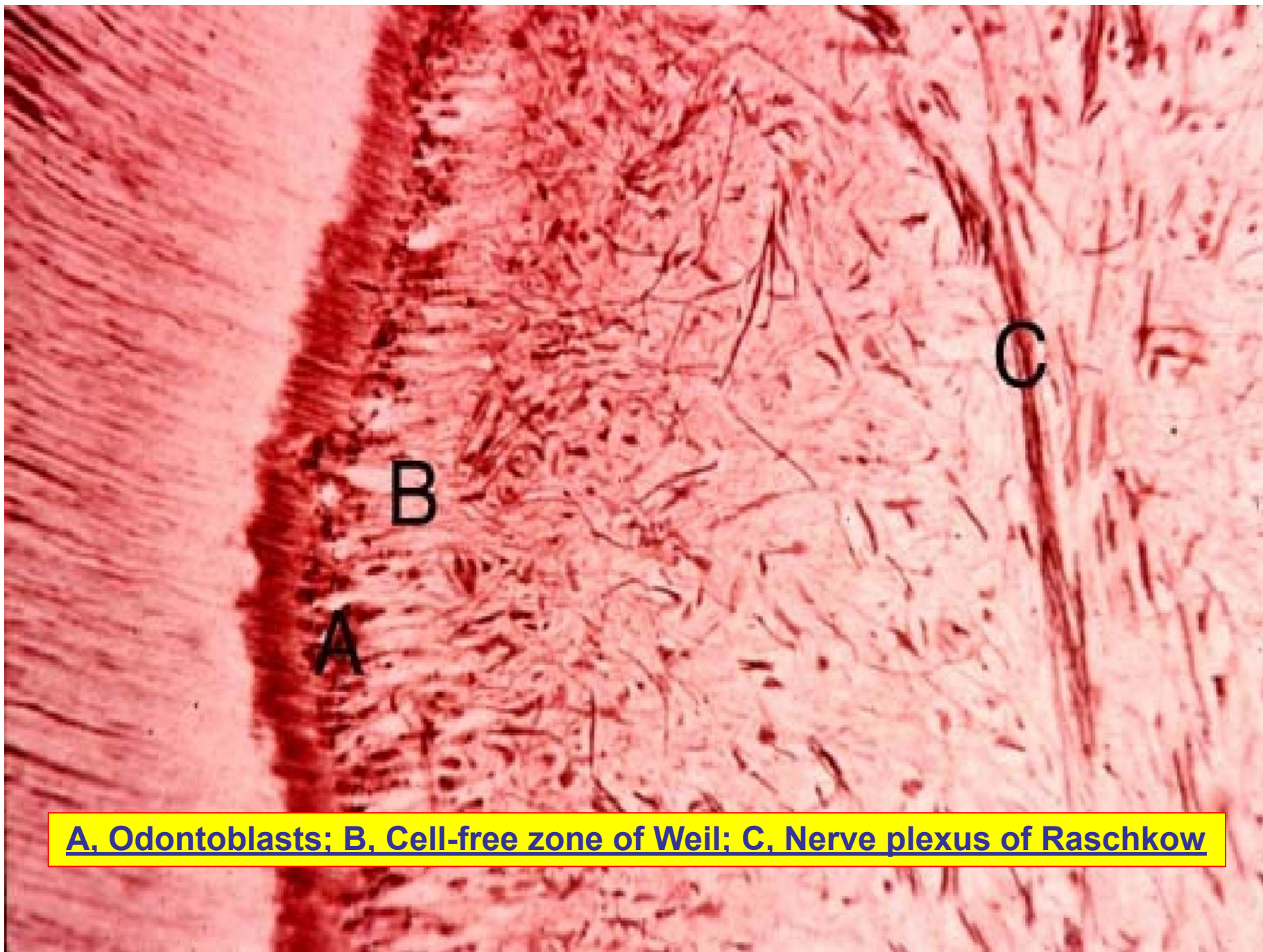


Fig. 10.39 Terminal branching of sensory fibres (arrows) in the dental pulp. This very thick section allows the axons at several levels to be seen, including the subodontoblastic nerve plexus of Raschkow (silver stain; $\times 140$).



A, Odontoblasts; B, Cell-free zone of Weil; C, Nerve plexus of Raschkow

A δ Fibres (Myelinated Nociceptors)

- All pulpal A δ fibres are considered **nociceptive**.
- Transmit **sharp, well-localized pain**.
- No convincing evidence that the pulp senses anything other than pain.
- **Pain:** Subjective sensory and emotional experience.
- **Nociception:** Neural process of detecting and transmitting noxious stimuli.

C Fibres (Unmyelinated Afferents)

- Conduct **slow, dull, aching pain**.
- Conduction velocity: **0.5–2 m/s**.
- Important in **axon reflexes**:
 - Impulse travels to neighboring branches.
 - Causes local release of neuropeptides (e.g., Substance P, CGRP).
 - Regulates pulpal blood flow and neurogenic inflammation.
- Possible trophic effects on odontoblasts have been proposed but remain **unconfirmed**.

Nociceptor Type	Stimulus	Fibre Type	Conduction Velocity
Thermal	>45°C or <5°C	A δ	5–30 m/s
Mechanical	Intense pressure	A δ	5–30 m/s
Polymodal	Mechanical, chemical, thermal	A δ / C	0.5–2 m/s

CENTRAL PATHWAY

- Trigeminal Ganglion → Trigeminal Sensory Root → Trigeminal Spinal Nucleus (Subnucleus Caudalis)
- **Nociceptive-specific neurons:** Signal location and presence of noxious stimuli.
- **Nociceptive-nonspecific neurons:** Integrate inputs from multiple receptors and encode stimulus intensity.
- Convergence and divergence in the **subnucleus caudalis** explain:
 - Diffuse/radiating pulpal pain
 - **Referred pain** to adjacent teeth or regions

NERVE ENDINGS

- Termination sites are difficult to map completely.
- Autonomic nerves terminate on arteriolar smooth muscle via specialized neuromuscular junctions.
- Some A δ sensory fibres extend into dentinal tubules (mainly coronal dentine and predentine).
- Other fibres terminate at the pulp–predentine junction (Raschkow's plexus), ideally positioned to detect dentinal stimuli.
- Many nerve endings lie close to odontoblasts; functional interactions are suggested by the presence of synapsin and synaptotagmin proteins, although true synapses have not been confirmed.
- Terminal fibres lose their Schwann cell sheath, leaving exposed axolemma that is highly sensitive to extracellular changes.
- These endings may trigger local neuropeptide release (axon reflexes) and contribute to pulpal defense and inflammation rather than direct sensation.

NEUROPEPTIDES

- Most significant: **CGRP (calcitonin-gene-related peptide)** — potent vasodilator, likely principal local blood flow regulator in pulp periphery.
1. Synthesised in trigeminal ganglion cell bodies, transported peripherally by axonal transport.
 2. Signal from pulp periphery (possibly NGF) modulates CGRP production at ganglion.
 3. CGRP may also stimulate BMP-2 production (dentine formation) in cultured pulp cells.
- NGF (nerve growth factor) — receptors on odontoblasts + pulp nerves; NGF itself donated by fibroblasts (not odontoblasts); increases with pulp injury; may chemoattract leukocytes.

Table 10.2 Neuropeptides and neurotransmitters identified in the dental pulp with possible roles (largely interpolated from actions in other tissues)

Neuropeptide/small molecule transmitter	Possible role
Calcitonin-gene-related peptide	Vasodilatation, stimulates cell division in pulpal fibroblasts
Substance P	Vasodilatation, ?nociceptive transmitter, stimulates cell division in pulpal fibroblasts
Neuropeptide Y	Sympathetic vasoconstrictor
Noradrenaline (norepinephrine)	Sympathetic vasoconstrictor
Enkephalin	Silencer of nociceptors
Somatostatin	Silencer of nociceptors
Endorphin	Silencer of nociceptors
Dopamine	Vasoactive or a precursor of adrenaline (epinephrine)
Adrenaline (epinephrine)	Vasoconstrictive via smooth muscle in arterioles
Cholecystokinin	Unknown
Vasoactive intestinal polypeptide	?Parasympathetic
Secretoneurin	Axon reflexes
Neurokinin A	Vasodilatation, ?nociceptive transmitter, stimulates cell division in pulpal fibroblasts
Peptide histidine isoleucine amide	?Parasympathetic
?Acetylcholine	?Parasympathetic

FUNCTIONS OF DENTAL PULP

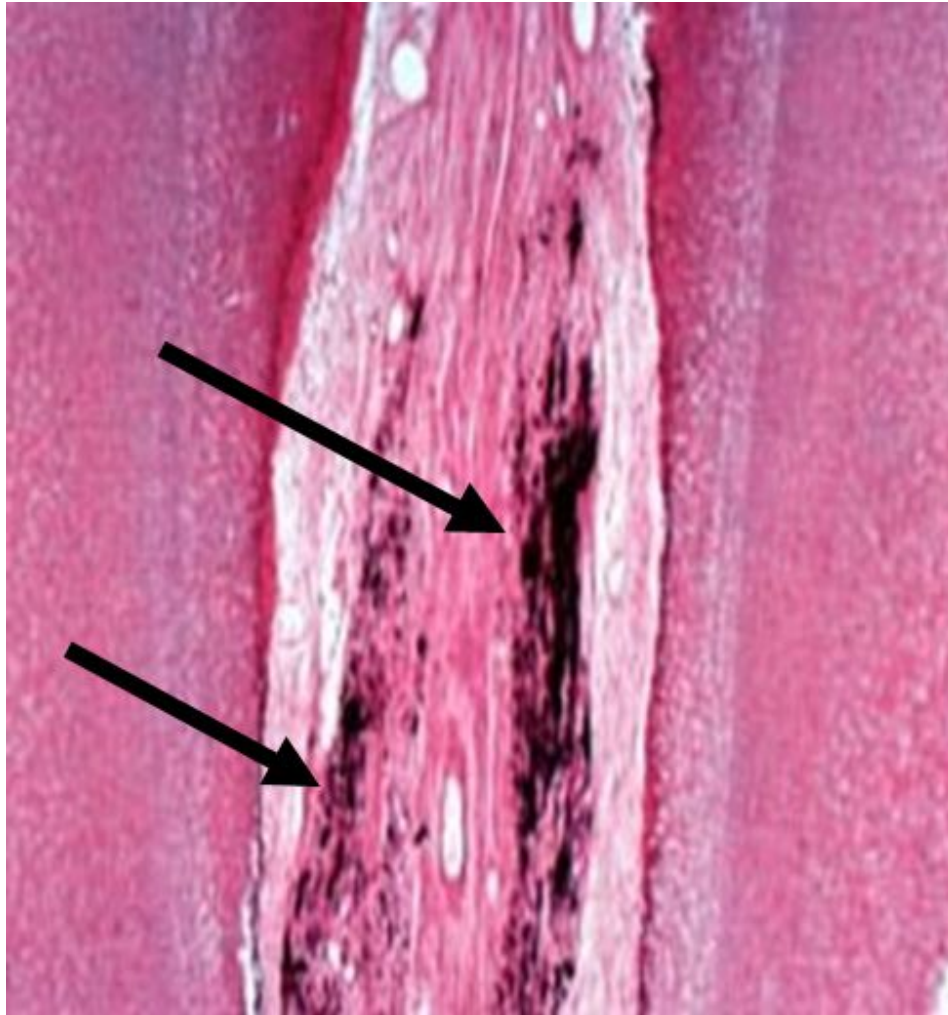
- **FORMATIVE**
- **NUTRITIVE**
- **SENSORY**
- **DEFENSIVE**

Function	Cells/Structures Involved	Description	Clinical Significance
Formative (Dentinogenic)	Odontoblasts	Forms primary, secondary, and tertiary (reparative) dentin throughout life.	Maintains dentin thickness and repairs damage caused by caries or trauma.
Nutritive	Blood vessels, capillaries	Supplies oxygen, nutrients, and metabolites to odontoblasts and dentin.	Essential for maintaining tooth vitality.
Sensory	A δ fibers, C fibers, nerve plexus of Raschkow	Transmits pain in response to thermal, mechanical, chemical, and electrical stimuli.	Responsible for tooth sensitivity and pulpal pain.

Function	Cells/Structures Involved	Description	Clinical Significance
Defensive (Protective)	Odontoblasts, macrophages, dendritic cells, lymphocytes	Responds to injury by producing tertiary dentin and initiating immune and inflammatory responses.	Protects the pulp from bacterial invasion and other irritants.

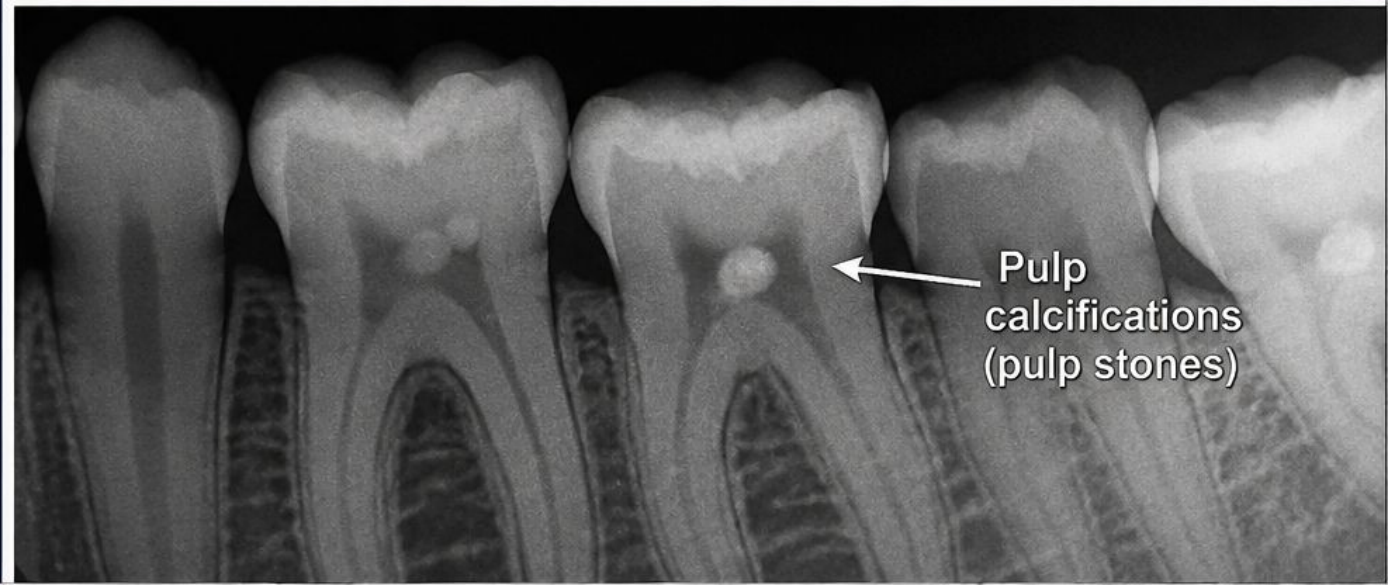
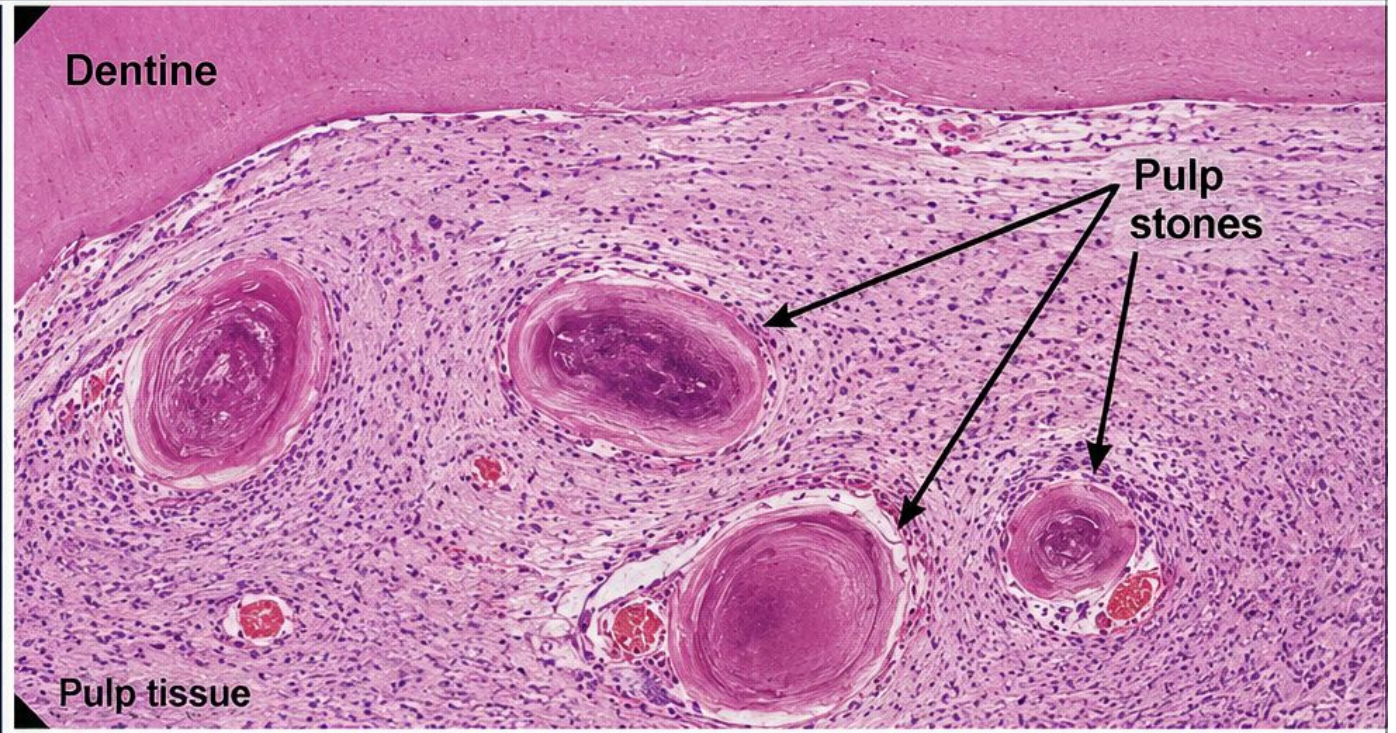
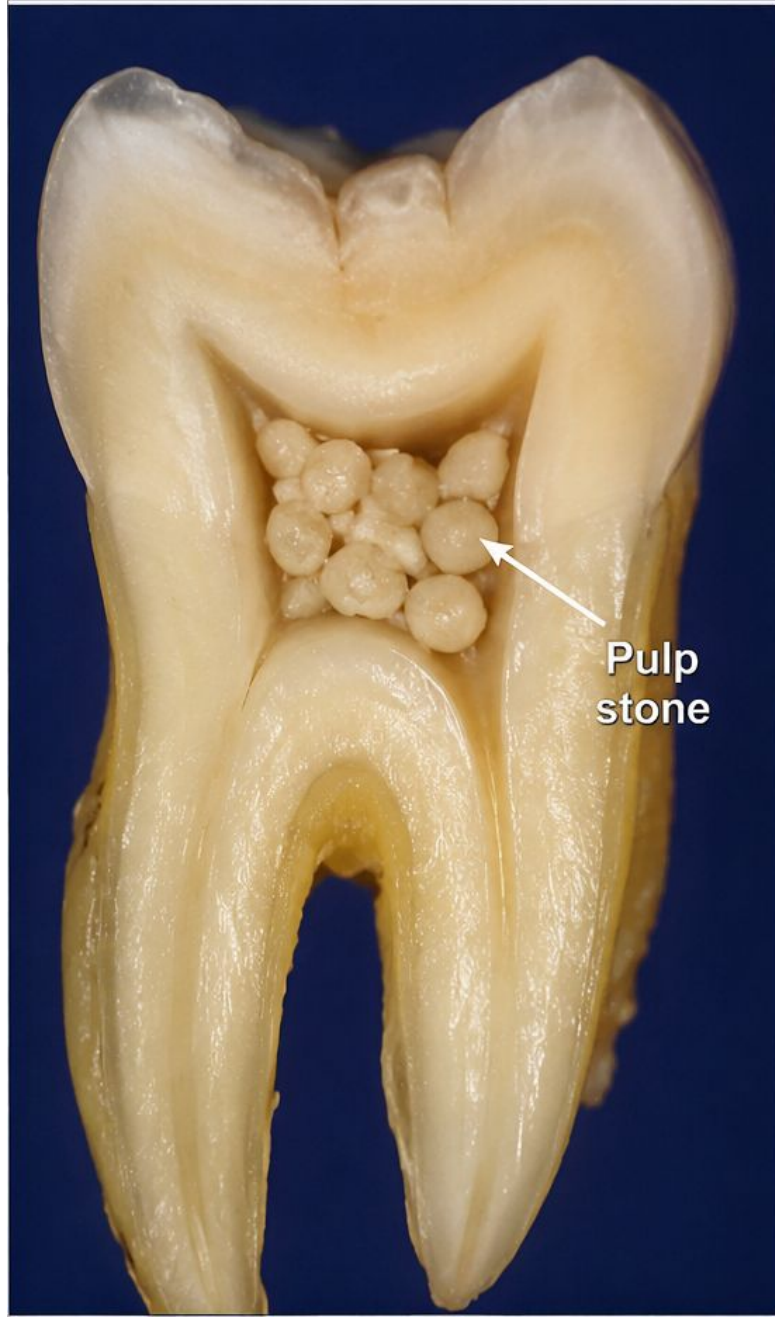
AGE RELATED CHANGES IN PULP

- Continuous **secondary dentin formation** reduces pulp size.
- Aging pulp shows:
 - ↓ Vascularity
 - ↑ Fibrosis
 - ↓ Innervation
- **Pulp Calcifications Types:**
- **Diffuse Snowstorm calcifications:** Numerous small calcified spicules
- **Pulp stones:** Discrete calcified masses



**DIFFUSED
CALCIFICATION**

PULP CALCIFICATION



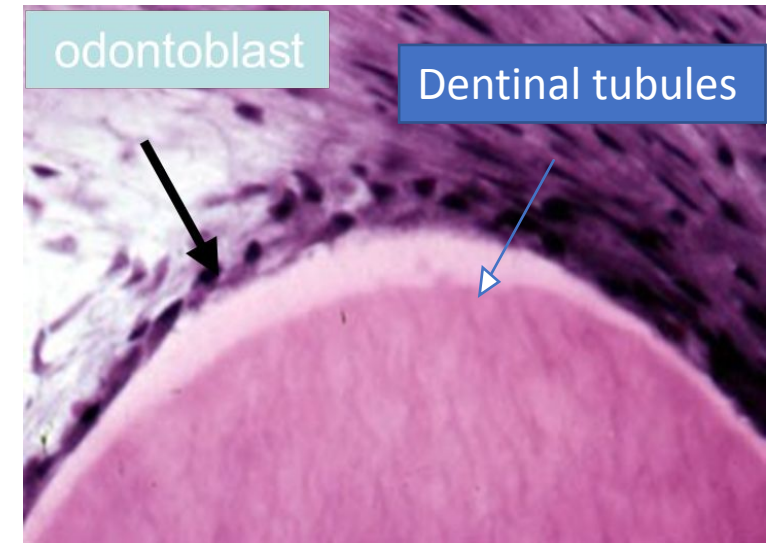
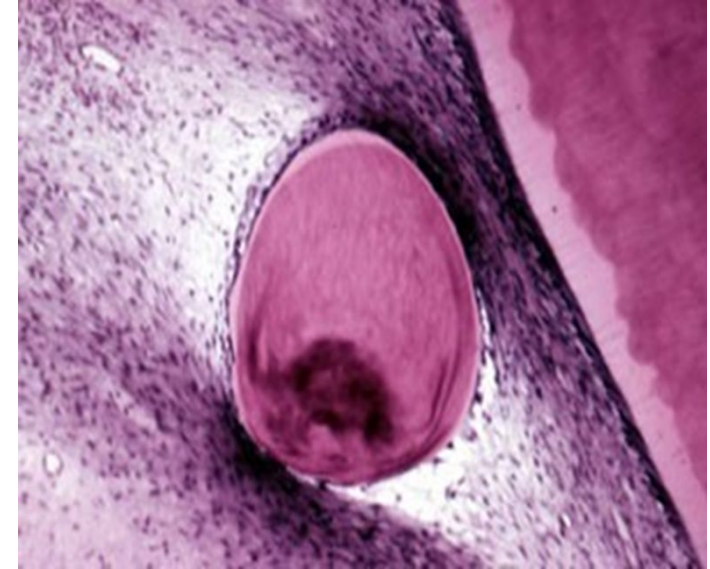
CONTD.

TYPE OF PULP STONES

- **True denticles:** Dentin-like with tubules
- **False denticles:** Bone-like, cell-containing
- **Lamellated pulp stones:** Layered calcifications
- Usually detected radiographically.
- Asymptomatic stones do not require RCT but may complicate treatment.

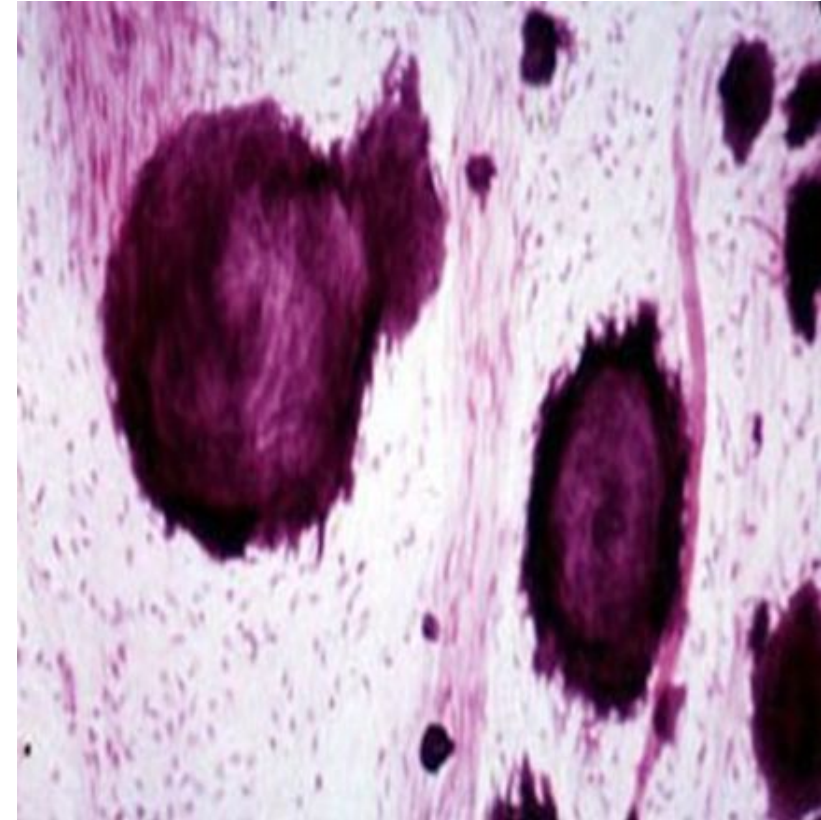
TRUE DENTICLES

- True denticles are rare & small in size & found near the apical foramen.
- They consist of irregular dentin containing traces of dentinal tubules and few odontoblasts.
- Remnants of the epithelial root sheath invade the pulp tissues causing Undifferentiated mesenchymal cells of the pulp to form this irregular type of dentin.



FALSE DENTICLES

- They are evidence of dystrophic calcification of the pulp tissue.
- They contain no dentinal tubules.
- They are formed of degenerated cells or areas of hemorrhage which act as a central nidus for calcification.
- Overdoses of vit. D, may favor the formation of numerous denticles.



CONTD.

- Pulp stones are also classified according to their location into: **free, attached and embedded.**
- They continue to increase in size and in certain cases they fill up the pulp chamber completely.
- The close proximity of pulp stones to blood vessels may cause atrophy of it.



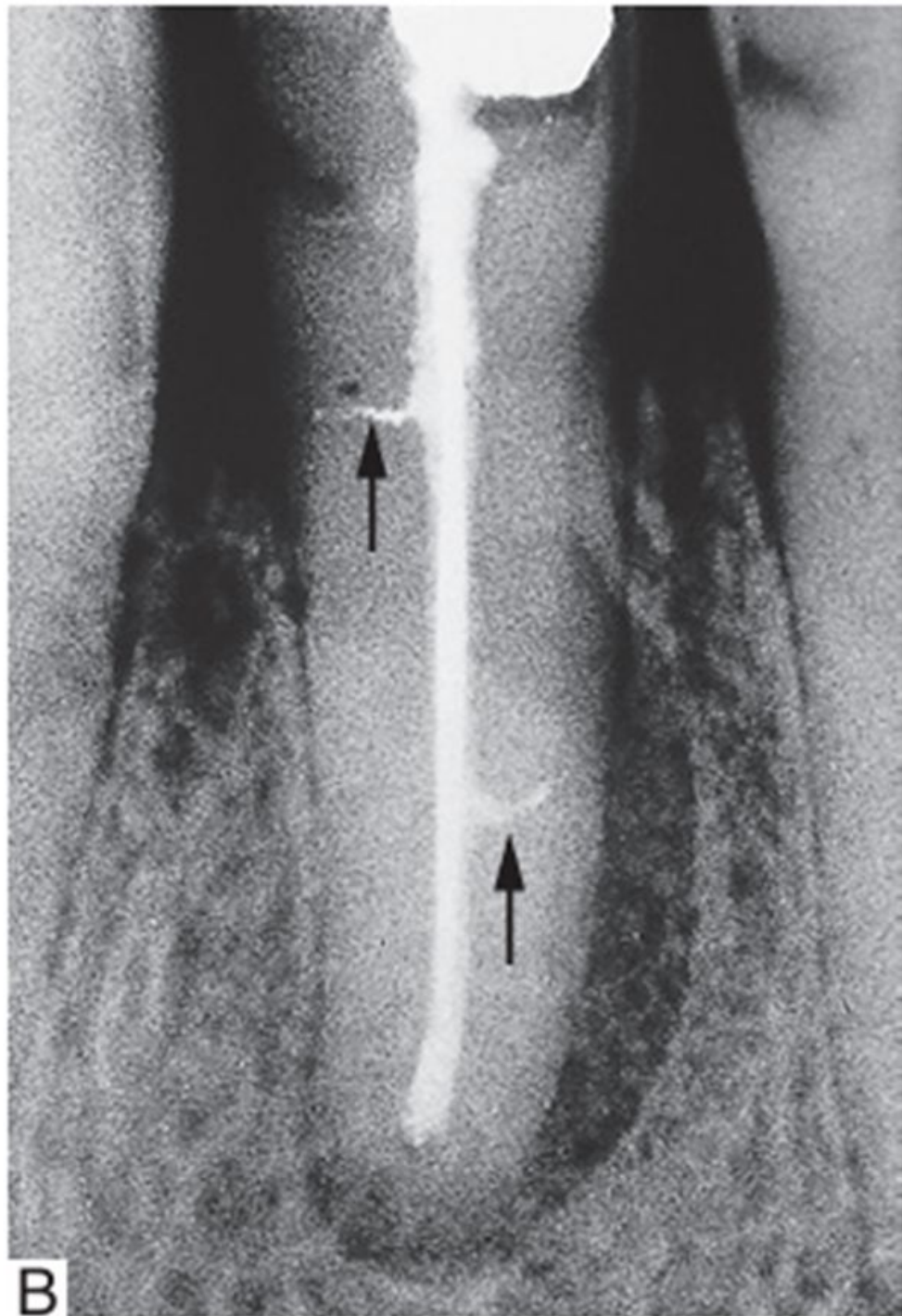
CLINICAL CONSIDERATIONS

Pulpal Response to Injury

- Pulp acts as a sensory and defensive organ.
- Can produce tertiary dentin after injury.
- Pulpitis may occur even with superficial caries.
- Pain is often difficult to localize
- Untreated inflammation may progress to Necrosis or Periapical disease

LATERAL ROOT CANALS

- Gingival recession can expose lateral canal openings (esp. furcation areas) → pain + infection spread risk into pulp.
- Periodontal disease may affect pulp and vice versa (endodontic periodontic lesions) — successful treatment depends on identifying origin of disease process



REGENERATION, APEXIFICATION AND REPAIR

PULP EXPOSURE TREATMENT

- An uninfected exposed pulp can, if treated appropriately, form a dentine bridge over the exposure via new odontoblast-like cells laying down tertiary (reparative) dentine.
- **Calcium hydroxide** facilitates bridge formation mainly via protection & biocompatibility.
- **Mineral trioxide aggregate (MTA)** may be more efficient, stimulating hard-tissue forming cells and forming apatite precipitates with phosphate fluids.
- If the exposed pulp is infected, bridge formation is much less likely and pulp removal + root canal filling is usually necessary

DIRECT PULP CAPPING



1. Pulp exposure
Mechanical exposure
with pinpoint bleeding



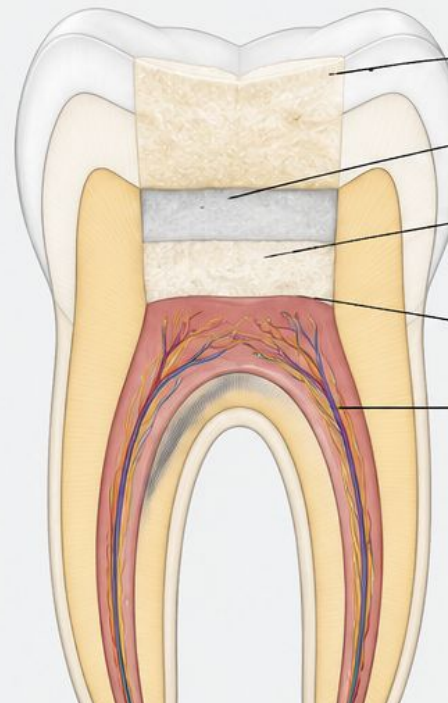
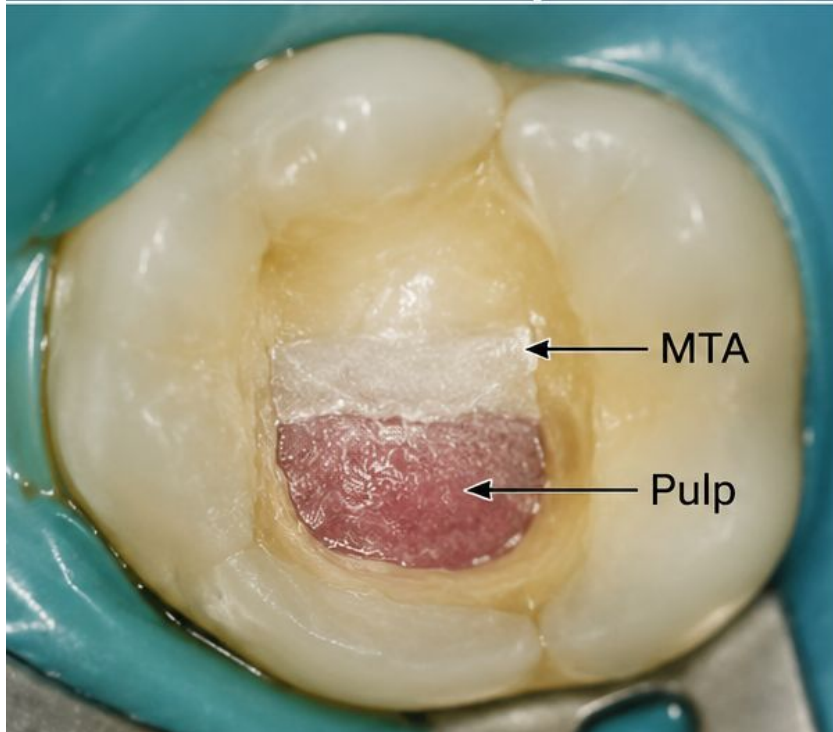
2. Medicament placement
Calcium hydroxide or MTA
placed over the exposure



3. Base placement
RMGIC (resin-modified
glass ionomer cement)



4. Final restoration
Tooth-colored composite
restoration placed



Composite restoration

RMGIC base

MTA / Ca(OH)_2
(pulp capping agent)

Dentin

Pulp
(vital)

OBJECTIVES

- Protect the exposed pulp
- Maintain pulp vitality
- Promote formation of tertiary dentin (dentin bridge)
- Prevent pulpal inflammation or necrosis

Tissue engineering

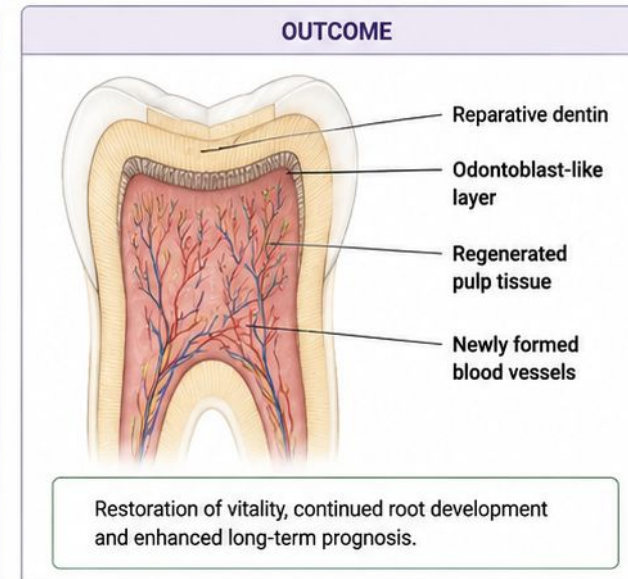
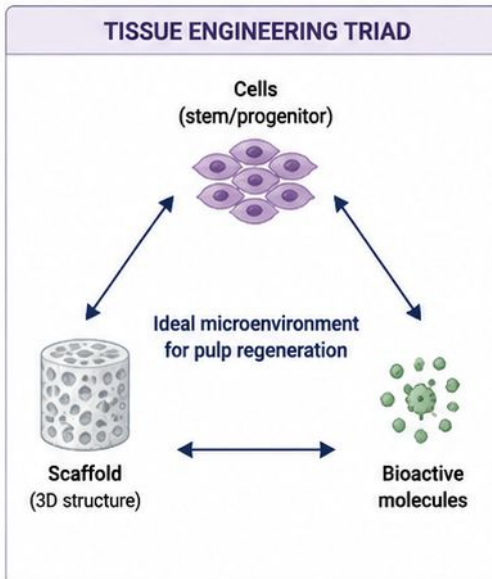
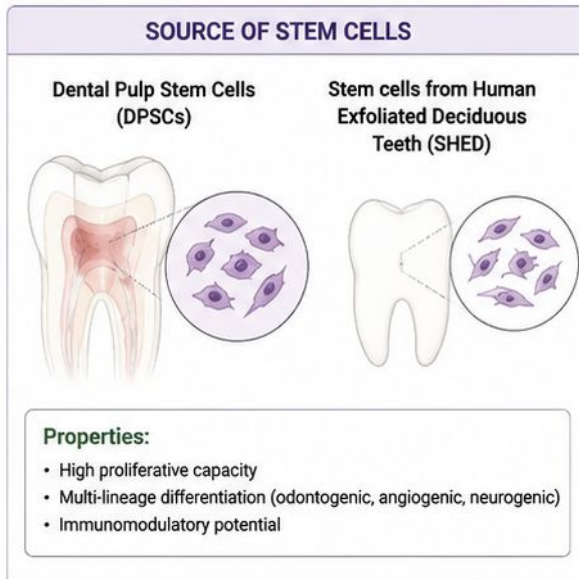
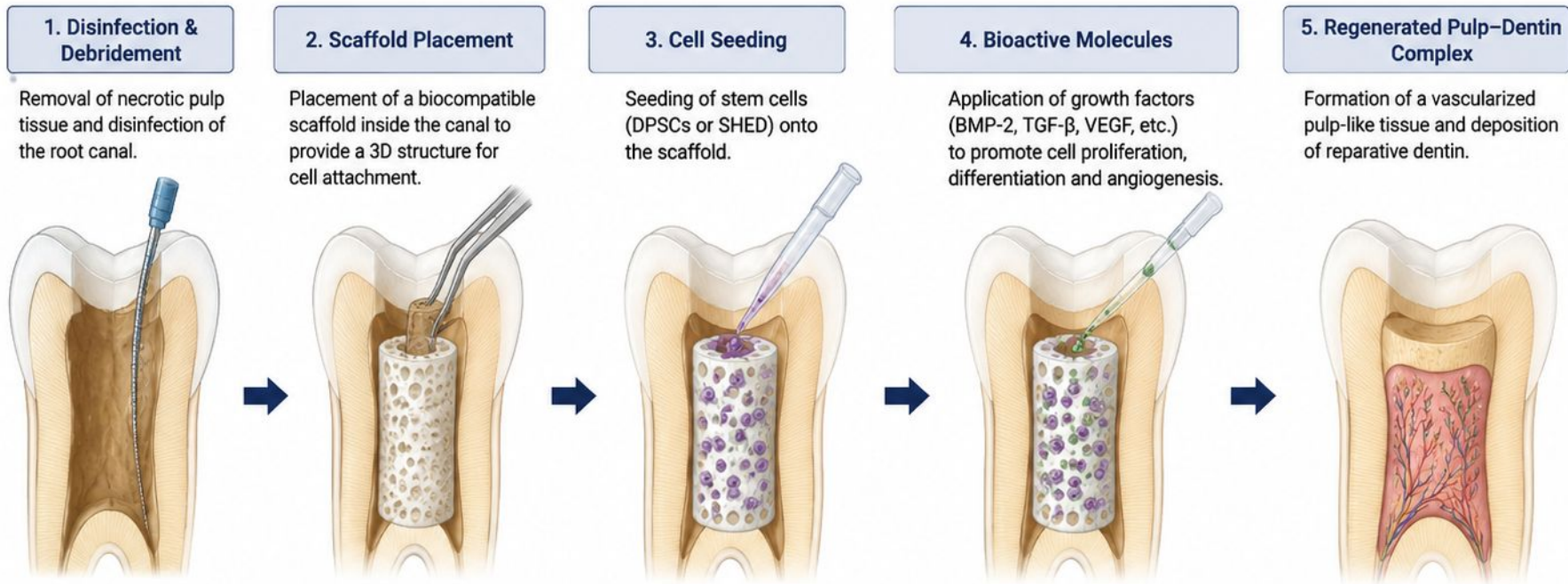
- Cell transplantation — fill root canal space with transplanted (ideally patient-derived) stem cells + scaffold + BMPs/FGFs
- Cell homing — fill with scaffold + signalling molecules that chemotactically recruit stem cells from the apex
- Proangiogenic factors (FGF2, VEGF) are important — a blood supply is crucial to either strategy's success.

Apexification and revascularization:

- For immature necrotic teeth: irrigate & remove necrotic tissue, seal with antibiotic paste, protect the apical stem-cell-rich region (SCAPs) with calcium hydroxide. Later, induce apical bleeding to form a blood clot — bringing in stem cells to complete a vital root apex.
- MTA can also seal the apex directly (apexification), depositing calcium phosphates as pH rises

TISSUE ENGINEERING OF DENTAL PULP

Regeneration of a vital pulp–dentin complex



REVASCULARIZATION / REGENERATION (Regenerative Endodontic Procedure – REP)

INDICATIONS

- Immature permanent tooth with necrotic pulp
- Open apex
- Periapical pathology

GOAL

To regenerate pulp–dentin complex and continue root development.

1. Disinfection



Cleaning and shaping with minimal instrumentation. Irrigation with NaOCl and EDTA. Intracanal medication (e.g., Triple antibiotic paste) for 2–4 weeks.

2. Induction of Bleeding



After disinfection, irrigate and dry. Induce bleeding beyond apex to form a blood clot up to cemento–enamel junction (CEJ).

3. Placement of Scaffold



Allow blood clot to act as natural scaffold.

4. Coronal Seal



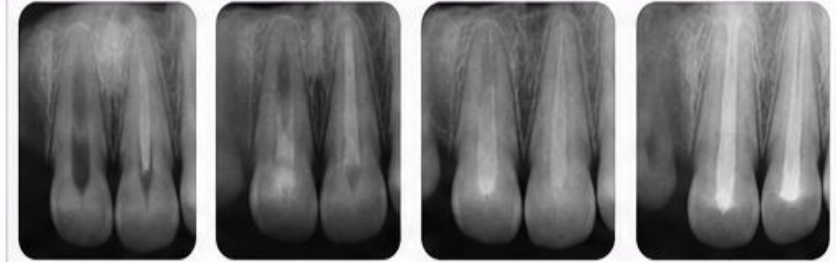
Place MTA (3–4 mm) over the clot and seal coronally with a permanent restoration.

5. Regeneration



Regeneration of pulp-like tissue, continued root development and apical closure.

RADIOGRAPHIC PROGRESSION



Pre-operative — 3 months — 6–12 months — 12–24 months →

POTENTIAL OUTCOMES

- ✓ Thickening of canal walls
- ✓ Increased root length
- ✓ Apical closure
- ✓ Resolution of periapical lesion
- ✓ Return of pulp vitality (not always)

KEY POINTS

- Uses stem cells from apical papilla, blood clot, and scaffold
- Growth factors released from clot stimulate regeneration
- Technique sensitive
- Long-term success improving

COMPARISON	Apexification	Revascularization / Regeneration
Tissue formed	Calcified (hard tissue) barrier	Vital pulp–like tissue
Root development	No further root development	Continued root development
Treatment time	Long (months to years)	Relatively shorter (months)
Biological potential	Barrier only	Complete regeneration potential

NOTE: Case selection, disinfection, and coronal seal are critical for success in both procedures.

PULP VITALITY TEST

Test	Mechanism	Key Point
Thermal test	Cold/heat stimulation of nerves	Cold test (tetrafluoroethane) is more reliable. Heated gutta percha can also be used.
Electrical pulp test	Electrical stimulation of nerve fibres	Indicates nerve presence
Pulp oximetry	Measures oxygen saturation	Most reliable; assesses blood supply

COLD TEST



HOT TEST



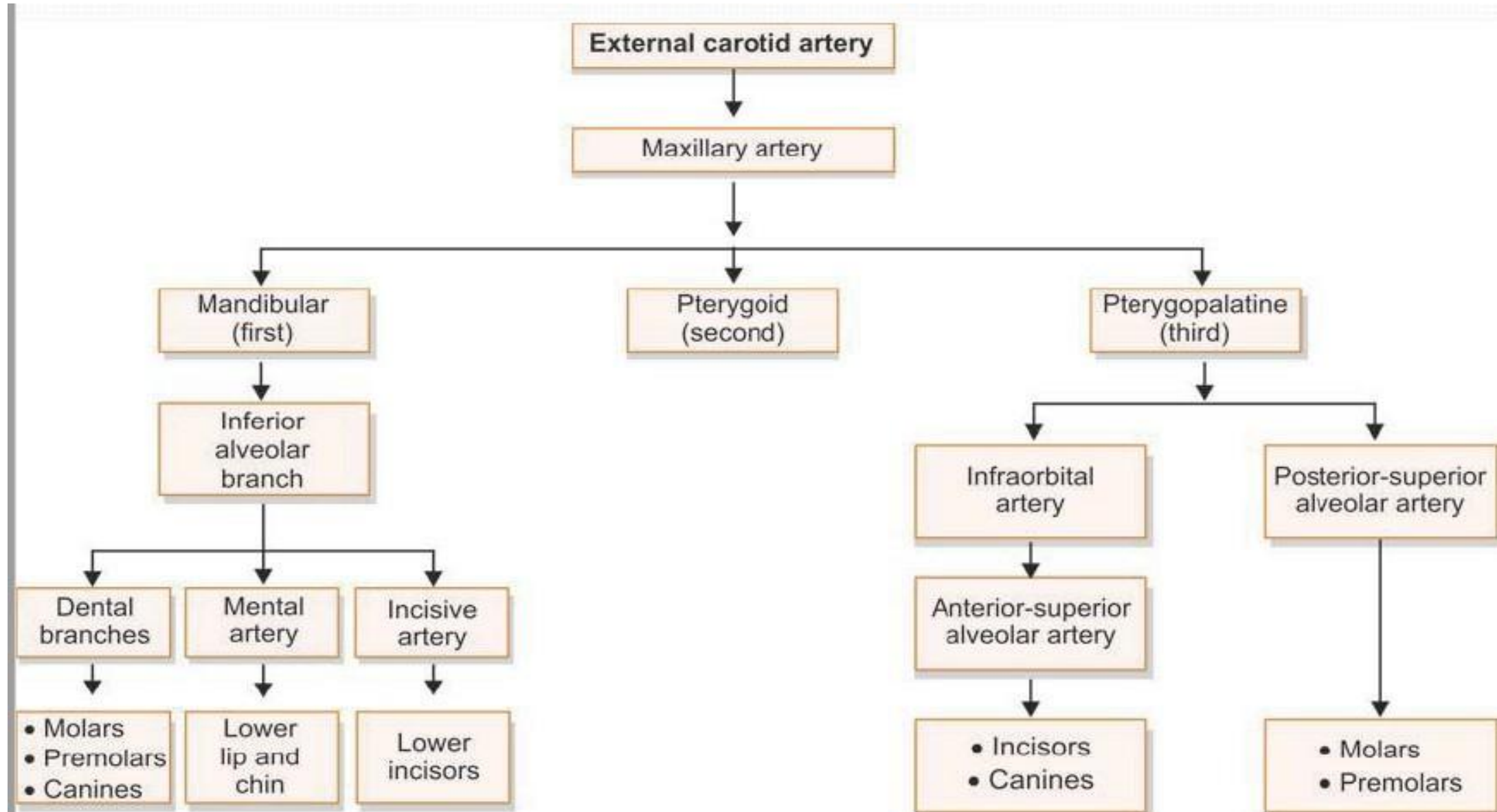
ELECTRIC PULP TEST



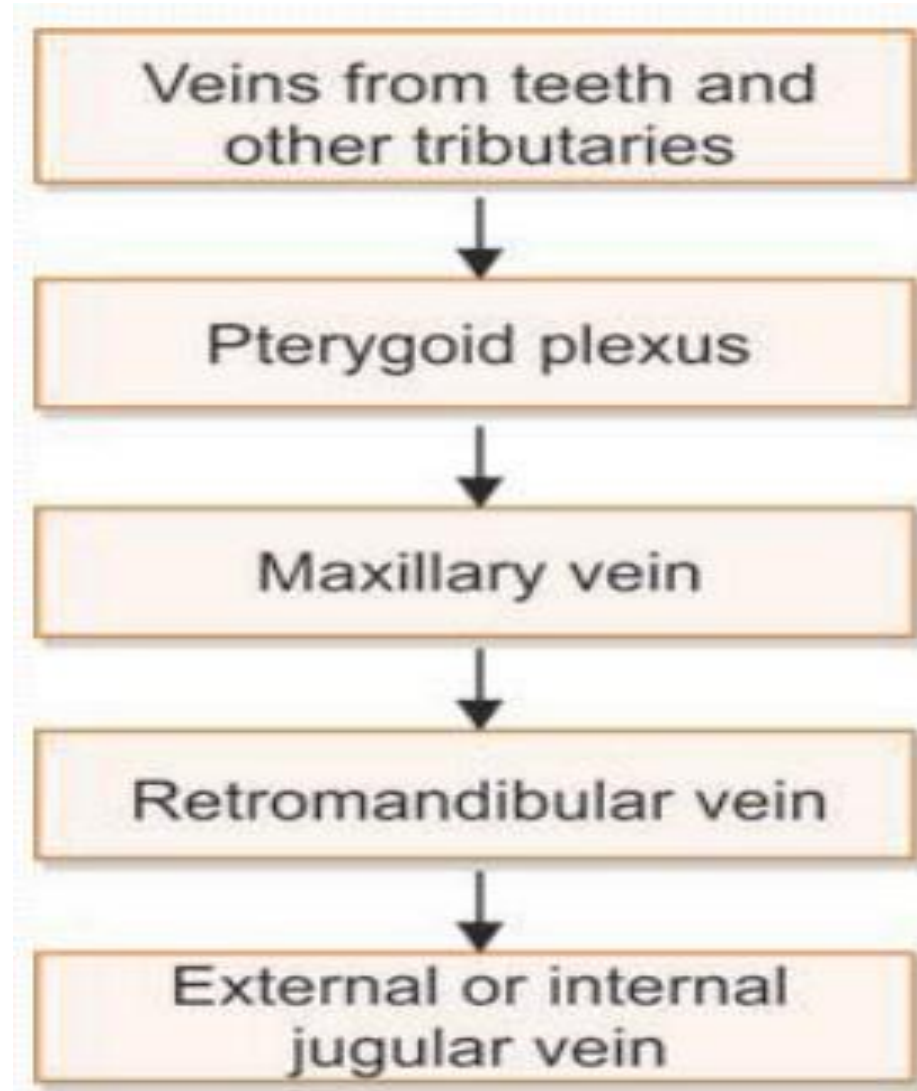
PULP OXIMETER



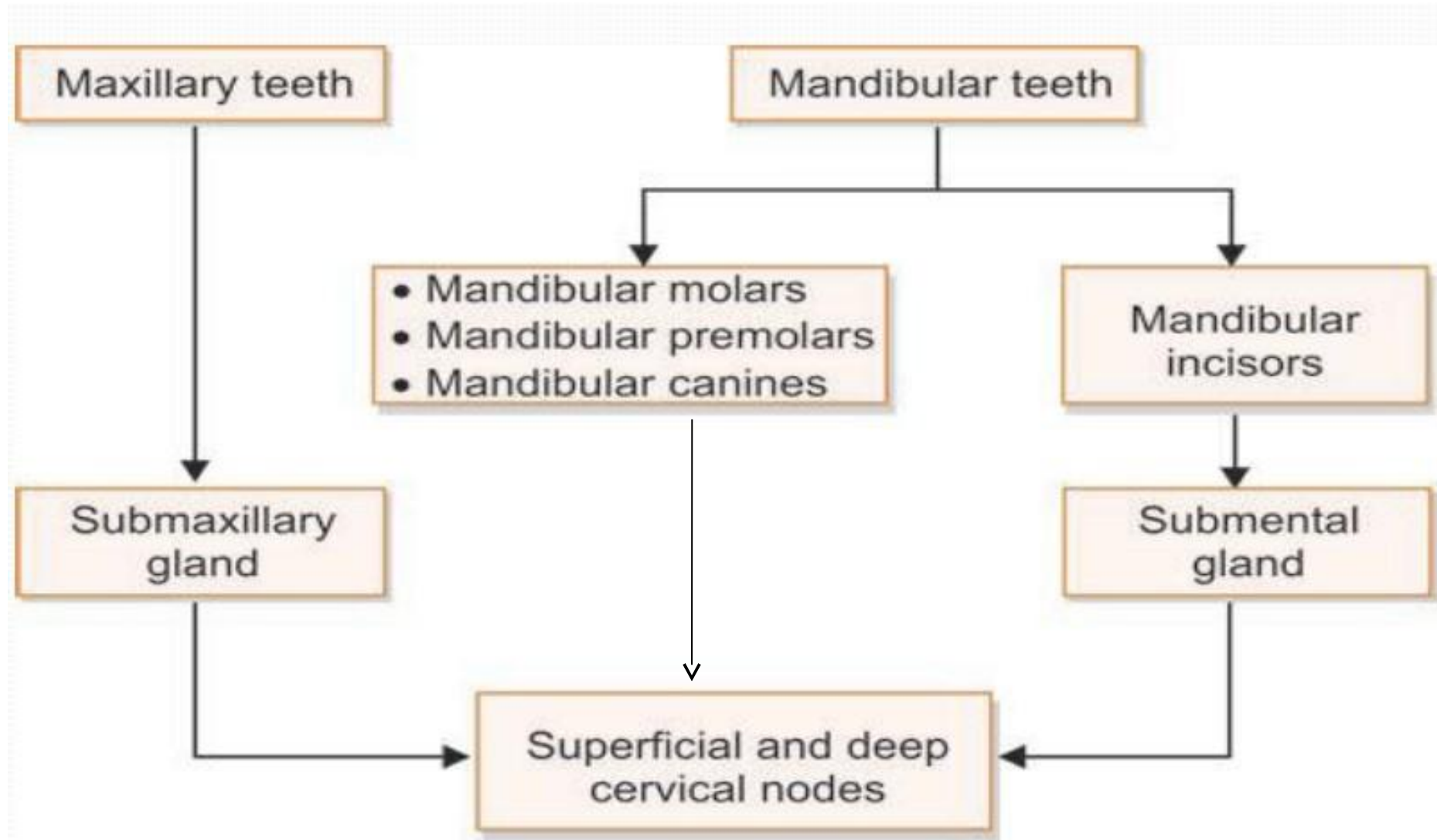
ARTERIAL SUPPLY OF TEETH



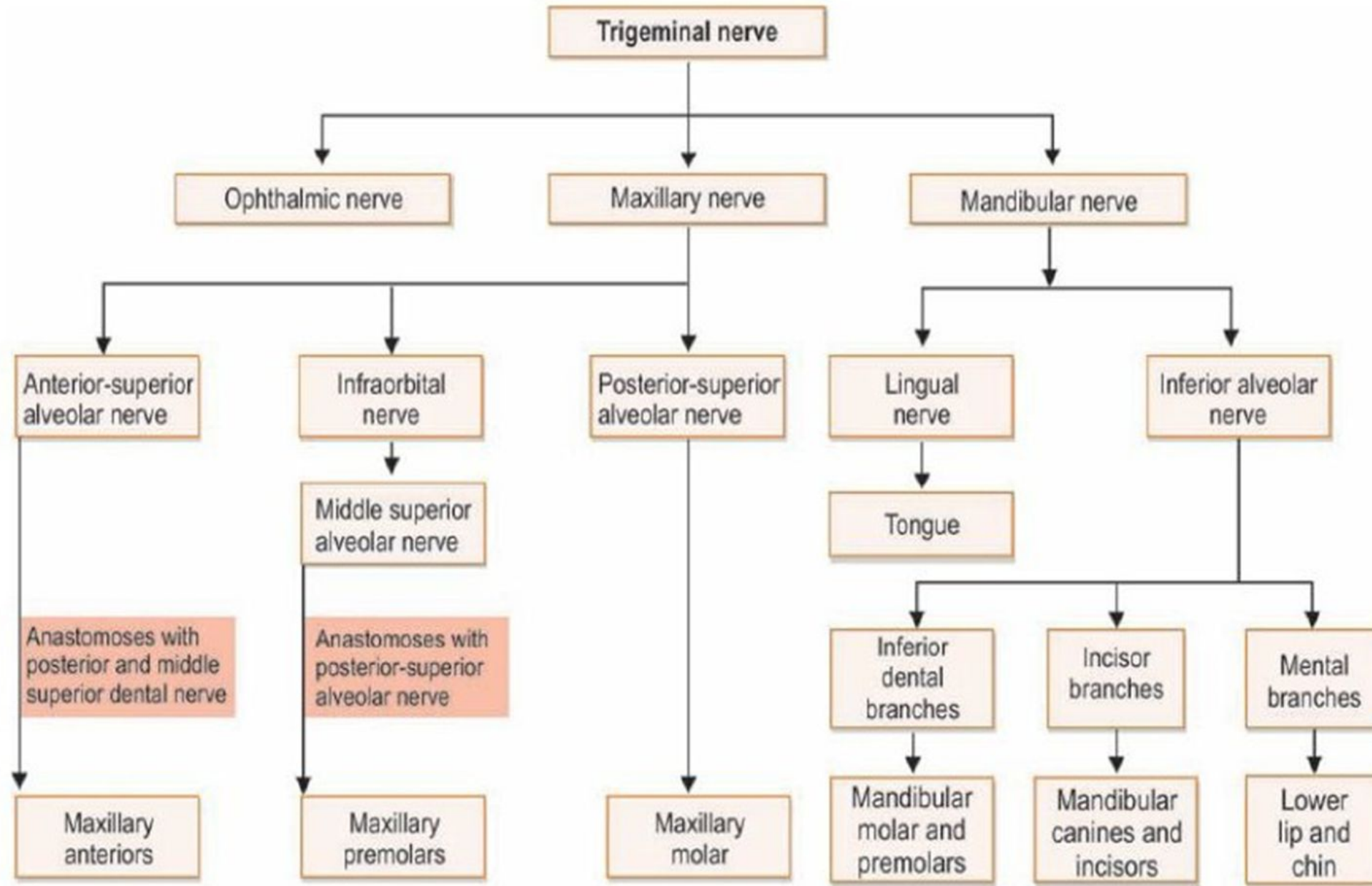
VENOUS SUPPLY OF TEETH



LYMPHATIC DRAINAGE OF TEETH



NERVE SUPPLY OF TEETH



KEY TAKEAWAYS

- A specialized loose connective tissue, almost entirely surrounded by mineralised dentine
- Collagen types I (60%) & III (40%) dominate a matrix rich in proteoglycans & GAGs
- Odontoblasts, fibroblasts, stem cells and defence cells each play distinct, coordinated roles
- Unusually rich blood supply (20–60 mL/min/100g) and dense sensory innervation (A δ , A β , C fibres)
- All pulpal stimuli are perceived as pain — reflecting the odontoblast layer's barrier & sensory functions
- Ages by shrinking, reduced vascularity/innervation, and mineralisation into pulp stones
- Clinically central to caries response, vitality testing, pulp capping, and regenerative endodontics

REFERENCES

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- TEN CATES